11. Low Power Consumption

11.1 Overview

This MCU has several functions for reducing power consumption, including switching of clock signals to reduce power consumption, BCLK output control, SDCLK output control, stopping modules, functions for low power consumption in normal operation, and transitions to low power consumption states.

Table 11.1 lists the specifications of low power consumption functions, and Table 11.2 lists the conditions to shift to low power consumption modes, states of the CPU and peripheral modules, and the method for release from each mode.

After a reset, this MCU enters the normal program execution state, but modules except for the DMAC, DTC, and RAM do not operate.

Table 11.1 Specifications of Low Power Consumption Functions

Item	Specification
Reducing power consumption by switching clock signals	The frequency division ratio is settable independently for the system clock (ICLK), peripheral module clock (PCLKA, PCLKB, PCLKC, PCLKD), external bus clock (BCLK), and flash interface clock (FCLK).*1
BCLK output control function	BCLK output or high-level output can be selected.*1
SDCLK output control function	SDCLK output or high-level output can be selected.*1
Module-stop function	Functions can be stopped independently for each peripheral module.
Function for transition to low power consumption mode	Transition to a low power consumption mode in which the CPU, peripheral modules, or oscillators are stopped is enabled.
Low power consumption modes	• Sleep mode • All-module clock stop mode • Software standby mode • Deep software standby mode
Function for lower operating power consumption	• Power consumption can be reduced in normal operation, sleep mode, and all-module clock stop mode by selecting an appropriate operating power consumption control mode according to the operating frequency and operating voltage range. • Three operating power control modes High-speed operating mode Low-speed operating mode 1 Low-speed operating mode 2 There is no difference in power consumption when the same conditions (frequency and voltage) are set in low-speed operating modes 1 and 2.

Note 1. For details, refer to section 9, Clock Generation Circuit.

Table 11.2 Entering and Exiting Low Power Consumption Modes and Operating States in Each Mode

Entering and Exiting Low Power Consumption Modes and Operating States	Sleep Mode	All-Module Clock Stop Mode	Software Standby Mode	Deep Software Standby Mode
Transition condition	Control register + instruction	Control register + instruction	Control register + instruction	Control register + instruction
Method of release other than reset	Interrupt	Interrupt*1	Interrupt*2	Interrupt*3
State after release*4	Program execution state (interrupt processing)	Program execution state (interrupt processing)	Program execution state (interrupt processing)	Program execution state (reset processing)
Main clock oscillator	Operating possible	Operating possible	Operating possible*5	Operating possible*5
Sub-clock oscillator	Operating possible	Operating possible	Operating possible*6	Operating possible*6
High-speed on-chip oscillator	Operating possible	Operating possible	Stopped	Stopped
Low-speed on-chip oscillator	Operating possible	Operating possible	Stopped	Stopped
IWDT-dedicated on-chip oscillator	Operating possible*7	Operating possible*7	Operating possible*7	Stopped (Undefined)*7
PLL	Operating possible	Operating possible	Stopped	Stopped
PPLL	Operating possible	Operating possible	Stopped	Stopped
CPU	Stopped (Retained)	Stopped (Retained)	Stopped (Retained)	Stopped (Undefined)
RAM, expansion RAM, and ECCRAM	Operating possible (Retained)	Stopped (Retained)	Stopped (Retained)	Stopped (Undefined)
Standby RAM	Operating possible (Retained)	Stopped (Retained)	Stopped (Retained)	Stopped (Retained/Undefined)*8
Flash memory	Operating	Stopped (Retained)	Stopped (Retained)	Stopped (Retained)
USBFS host/function module (USB)	Operating possible	Stopped*9	Stopped*9	Stopped (Retained/Undefined)*10
Watchdog timer (WDT)	Stopped (Retained)	Stopped (Retained)	Stopped (Retained)	Stopped (Undefined)
Independent watchdog timer (IWDT)	Operating possible*7	Operating possible*7	Operating possible*7	Stopped (Undefined)*7
Realtime clock (RTC)	Operating possible	Operating possible	Operating possible	Operating possible
8-bit timer (unit 0, unit 1) (TMR)	Operating possible	Operating possible*11	Stopped (Retained)	Stopped (Undefined)
Port output enable (POE)	Operating possible	Operating possible*16	Stopped (Retained)	Stopped (Undefined)
Voltage detection circuit (LVD)	Operating possible	Operating possible	Operating possible	Operating possible *12, *13
Power-on reset circuit	Operating	Operating	Operating	Operating*13
Peripheral modules	Operating possible	Stopped (Retained)	Stopped (Retained)	Stopped (Undefined)
I/O ports	Operating	Retained*14	Retained*15	Retained*15

"Operating possible" means that operating or stopped can be controlled by the control register setting.
"Stopped (Retained)" means that internal register values are retained and internal operations are suspended.
"Stopped (Undefined)" means that internal register values are undefined and power is not supplied to the internal circuit.

Note 1. "Interrupts" here indicates an external pin interrupt (the NMI or IRQ0 to IRQ15) or any of peripheral interrupts (the 8-bit timer, RTC alarm, RTC periodic, IWDT, USB suspend/resume, voltage monitoring 1, voltage monitoring 2, and main-clock oscillation stop detection).
Note 2. "Interrupts" here indicates an external pin interrupt (the NMI or IRQ0 to IRQ15) or any of peripheral interrupts (the RTC alarm, RTC periodic, IWDT, USB suspend/resume, voltage monitoring 1, and voltage monitoring 2 interrupts).
Note 3. "Interrupts" here indicates a certain external pin interrupt source pin (the NMI, IRQ0-DS to IRQ15-DS, SCL2-DS, SDA2-DS, or CRX1-DS) or any of peripheral interrupts (the RTC alarm, RTC periodic, USB suspend/resume, voltage monitoring 1, and voltage monitoring 2 interrupts). However, these interrupts are enabled only when the corresponding bit in the deep standby interrupt enable registers i (DPSIERi) (i = 0 to 3) is set to 1. When the pin functions have "-DS" appended to their names, they can also be used as triggers for release from deep software standby. Also, USB is not released from deep software standby mode using USB0_OVRCURB multiplexed with pin P22.
Note 4. This does not include release initiated by the RES# pin reset, power-on reset, voltage monitoring reset, or independent watchdog-timer reset. The transition is to the reset state when release is initiated by one of these reset sources.
Note 5. Operation or stopping can be selected by the main clock oscillator forced oscillation bit (MOFXIN) in the main clock oscillator forced oscillation control register (MOFCR).
Note 6. Operation or stopping is selected by the sub-clock oscillator control bit (RTCEN) in the RTC control register 3 (RCR3).
Note 7. Operation or stopping is selected by the setting of the IWDT sleep mode count stop control bit (IWDTSLCSTP) in the option function select register 0 (OFS0) in IWDT auto start mode. If the OFS0.IWDTSLCSTP bit is 0 (disabling stopping of the counter when a transition to low power consumption mode is made), the transition is to software standby mode rather than deep software standby mode. In any mode other than IWDT auto start mode, operation or stopping is selected by the setting of the sleep mode counter stop control bit (SLCSTP) in the IWDT counter stop control register (IWDTCSTPR). If the IWDTCSTPR.SLCSTP bit is 0 (disabling stopping of the counter when a transition to low power consumption mode is made),

the transition is to software standby mode rather than deep software standby mode.

Note 8. Retention or undefined is selectable by the setting of the deep cut bits in the deep standby control register (DPSBYCR.DEEPCUT[1:0]).

Note 9. Detection of USB resumption is possible.

Note 10. Disabling or enabling of detection of USB resumption is controllable by the deep cut bits in the deep standby control register (DPSBYCR.DEEPCUT[1:0]). When detection of USB resumption is enabled, the values of the registers in the USB resume detecting unit are only held even in deep software standby mode.
Also, USB is not released from deep software standby mode using USB0_OVRCURB multiplexed with pin P22.

Note 11. Stopping or operation is controlled by the module-stop setting bits (MSTPA4 and MSTPA5, respectively) in module-stop control register A (MSTPCRA) for 8-bit timers 0 and 1 (unit 0) and 2 and 3 (unit 1).

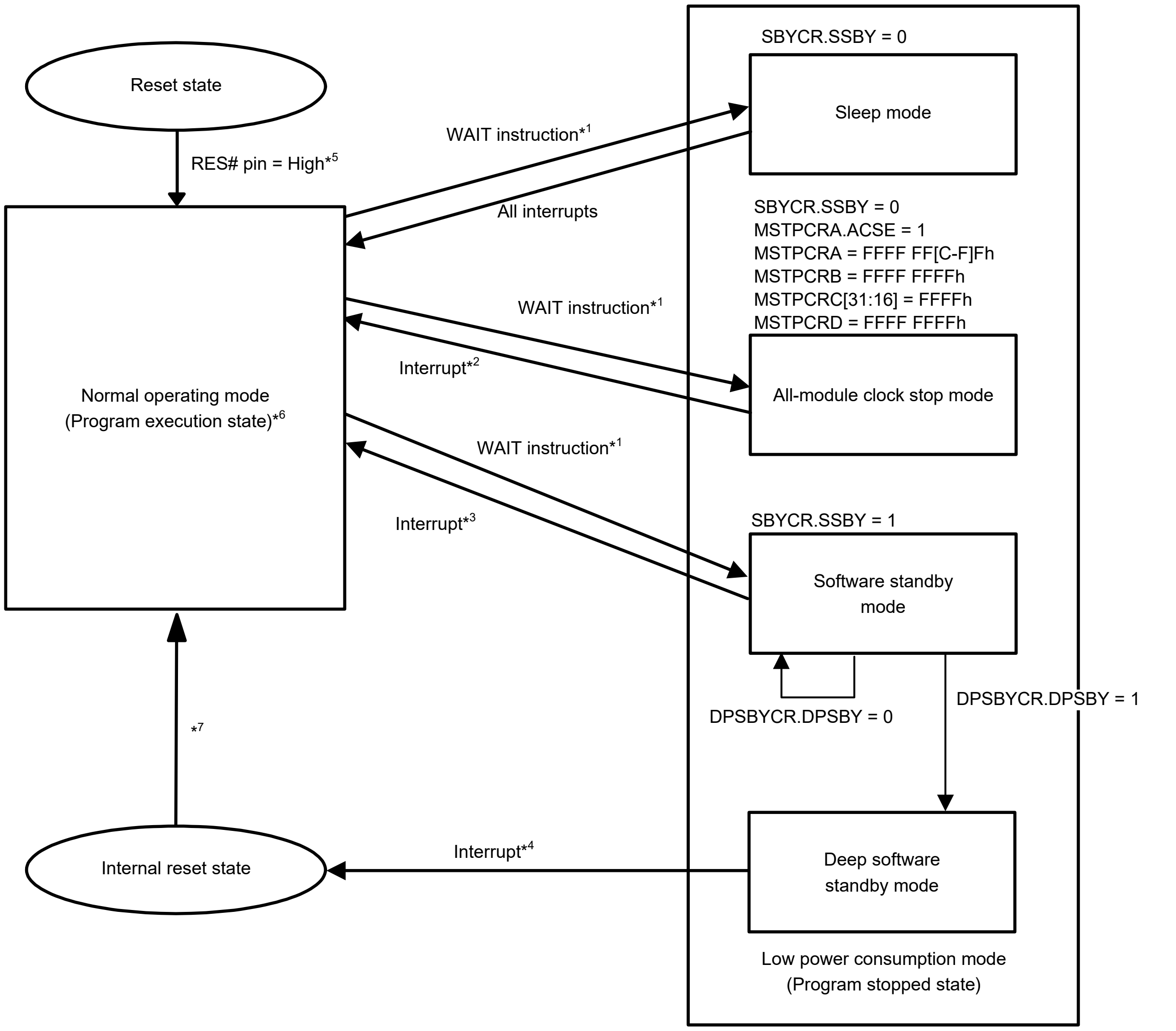
Note 12. If the voltage monitoring 1 circuit mode selection bit in the voltage monitoring 1 circuit control register 0 (LVD1CR0.LVD1RI) or the voltage monitoring 2 circuit mode selection bit in the voltage monitoring 2 circuit control register 0 (LVD2CR0.LVD2RI) is 1, the transition is to software standby mode rather than deep software standby mode.

Note 13. When the deep cut bits in the deep standby control register (DPSBYCR.DEEPCUT[1:0]) are set to 11b and the MCU enters deep software standby mode, the voltage detection circuit stops and the low power consumption function of the power-on reset circuit is enabled.

Note 14. If pin P53 is being used for the BCLK signal, operation continues with as-is output of BCLK. While the 8-bit timer and RTC are operated, the related pins continue operation.

Note 15. Retention of levels or placement in the high-impedance state is selectable for the address bus and bus control signals (CS0# to CS7#, RD#, WR0# to WR3#, WR#, BC0# to BC3#, ALE, CKE, SDCS#, RAS#, CAS#, WE#, and DQM0 to DQM3) by the output port enable bit (OPE) in the standby control register (SBYCR).

Note 16. When a source condition for POE interrupts is satisfied while POE interrupts are enabled and the chip is in all-module clock stop mode, the flag for the source condition is retained but return from all-module clock stop mode does not proceed. If a different source initiates return from all-module clock stop mode in this situation, the POE interrupt is generated after that.



Note 1. When an interrupt that acts as a trigger for release is received during the transition to the program-stopped state after the execution of a WAIT instruction, the transition to the program stopped state does not proceed and interrupt exception processing is executed instead.

Note 2. "Interrupts" here indicates an external pin interrupt (the NMI or IRQ0 to IRQ15) or any of peripheral interrupts (the 8-bit timer, RTC alarm, RTC periodic, IWDT, USB suspend/resume, voltage-monitoring 1, voltage-monitoring 2, and oscillator-stopped detection interrupts). However, a, 8-bit timer interrupt is only effective if the value of the corresponding module-stop setting bit (MSTPA4 or MSTPA5, respectively) in module-stop control register A (MSTPCRA) for 8-bit timer 0 and 1 (unit 0) or 2 and 3 (unit 1) is 0.

Note 3. "Interrupts" here indicates an external pin interrupt (the NMI or IRQ0 to IRQ15) or any of peripheral interrupts (the RTC alarm, RTC periodic, IWDT, USB suspend/resume, voltage-monitoring 1, and voltage-monitoring 2 interrupts).

Note 4. "Interrupts" here indicates a certain external pin interrupt source pin (the NMI, IRQ0-DS to IRQ15-DS, SCL2-DS, SDA2-DS, or CRX1-DS) or any of peripheral interrupts (the RTC alarm, RTC periodic, USB suspend/resume, voltage-monitoring 1, and voltage-monitoring 2 interrupts). However, these interrupts are enabled only when the corresponding bit in the deep standby interrupt enable registers i (DPSIERi) (i = 0 to 3) is set to 1. Also, USB is not released from deep software standby mode using USB0_OVRCURB multiplexed with pin P22.

Note 5. The LOCO is the source of the operating clock following a transition from the reset state to normal operating mode.

Note 6. The transition to normal operating mode is made due to an interrupt from sleep mode, all-module clock-stop mode, or software standby mode. In the case of recovery from sleep mode, the clock source for subsequent use is selectable. For details, refer to section 11.2.7, Sleep Mode Return Clock Source Switching Register (RSTCKCR). Also in the case of recovery from software standby mode, the clock source will be the same before and at the time of the transition.

Note 7. When an interrupt source as indicated in Note 4 above is generated, an internal reset (deep software standby reset) is generated over a fixed period. Release from deep software standby mode accompanies release from the internal reset state. The transition from deep software standby mode is to normal operating mode, and reset exception processing starts with the LOCO as the source of the operating clock.

When the RES# pin reset, power-on reset, voltage-monitoring 0 reset, voltage-monitoring 1 reset, voltage-monitoring 2 reset, watchdog timer reset, independent watchdog timer reset, or software reset is generated in any state, a transition to the reset state is made. The voltage-monitoring 1 reset and voltage-monitoring 2 reset cannot be used for release from deep software standby mode.

Figure 11.1 Mode Transitions

11.2 Register Descriptions

11.2.1 Standby Control Register (SBYCR)

Address(es): 0008 000Ch

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	SSBY	OPE	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b13 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b14	OPE	Output Port Enable	0: In software standby mode or deep software standby mode, the address bus and bus control signals are set to the high-impedance state. 1: In software standby mode or deep software standby mode, the address bus and bus control signals retain the output state.	R/W
b15	SSBY	Software Standby	0: Shifts to sleep mode or all-module clock stop mode after the WAIT instruction is executed 1: Shifts to software standby mode after the WAIT instruction is executed	R/W

OPE Bit (Output Port Enable)

The OPE bit specifies whether to retain the output of the address bus and bus control signals (CS0# to CS7#, RD#, WR0# to WR3#, WR#, BC0# to BC3#, ALE, CKE, SDCS#, RAS#, CAS#, WE#, and DQM0 to DQM3) in software standby mode or deep software standby mode, or to set the output to the high-impedance state.

SSBY Bit (Software Standby)

The SSBY bit specifies the transition destination after the WAIT instruction is executed.
When the SSBY bit is set to 1, the MCU enters software standby mode after execution of the WAIT instruction. When the MCU returns to normal operating mode after an interrupt has initiated release from software standby mode, the SSBY bit remains 1. Write 0 to this bit to clear it.
When the oscillation stop detection function enable bit in the oscillation stop detection control register (OSTDCR.OSTDE) is 1, the setting of the SSBY bit is ineffective. Even if the SSBY bit is 1, the MCU will enter sleep mode or all module clock stop mode on execution of the WAIT instruction.
When the code flash memory P/E mode entry bit (FENTRYR.FENTRYC) in the flash P/E mode entry register is 1 or the data flash memory P/E mode entry bit (FENTRYR.FENTRYD) is 1, the setting of this bit is ineffective. Sleep mode is entered on execution of the WAIT instruction even if this bit has been set to 1.

11.2.2 Module Stop Control Register A (MSTPCRA)

Address(es): 0008 0010h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	ACSE	—	MSTPA 29	MSTPA 28	MSTPA 27	—	—	MSTPA 24	—	—	—	—	MSTPA 19	—	MSTPA 17	MSTPA 16
Value after reset:	0	1	0	0	0	1	1	0	1	1	1	1	1	1	1	1

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	MSTPA 15	MSTPA 14	MSTPA 13	—	MSTPA 11	MSTPA 10	MSTPA 9	—	MSTPA 7	—	MSTPA 5	MSTPA 4	—	—	MSTPA 1	MSTPA 0
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Bit Name	Description	R/W
b0	MSTPA0	Compare Match Timer W (Unit 1) Module Stop	Target module: CMTW unit 1 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b1	MSTPA1	Compare Match Timer W (Unit 0) Module Stop	Target module: CMTW unit 0 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b3, b2	—	Reserved	These bits are read as 1. The write value should be 1.	R/W
b4	MSTPA4	8-Bit Timer 3/2 (Unit 1) Module Stop	Target module: TMR3/TMR2 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b5	MSTPA5	8-Bit Timer 1/0 (Unit 0) Module Stop	Target module: TMR1/TMR0 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b6	—	Reserved	This bit is read as 1. The write value should be 1.	R/W
b7	MSTPA7	General PWM Timer/GPTW Dedicated Port Output Enable Module Stop	Target module: GPTW/POEG 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b8	—	Reserved	This bit is read as 1. The write value should be 1.	R/W
b9	MSTPA9	Multifunction Timer Pulse Unit Module Stop	Target module: MTU 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b10	MSTPA10	Programmable Pulse Generator (Unit 1) Module Stop	Target module: PPG1 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b11	MSTPA11	Programmable Pulse Generator (Unit 0) Module Stop	Target module: PPG0 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b12	—	Reserved	This bit is read as 1. The write value should be 1.	R/W
b13	MSTPA13	16-Bit Timer Pulse Unit 0 (Unit 0) Module Stop	Target module: TPU unit 0 (TPU0 to TPU5) 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b14	MSTPA14	Compare Match Timer (Unit 1) Module Stop	Target module: CMT unit 1 (CMT2, CMT3) 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b15	MSTPA15	Compare Match Timer (Unit 0) Module Stop	Target module: CMT unit 0 (CMT0, CMT1) 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b16	MSTPA16	12-bit A/D Converter (Unit 1) Module Stop	Target module: S12AD unit 1 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W

Bit	Symbol	Bit Name	Description	R/W
b17	MSTPA17	12-bit A/D Converter (Unit 0) Module Stop	Target module: S12AD unit 0 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b18	—	Reserved	This bit is read as 1. The write value should be 1.	R/W
b19	MSTPA19	12-bit D/A Converter Module Stop	Target module: 12-bit D/A 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b23 to b20	—	Reserved	These bits are read as 1. The write value should be 1.	R/W
b24	MSTPA24	Module Stop A24	Writing to and reading from this bit is enabled. When a transition to all-module clock stop mode is made, be sure that 1 has been written to this bit.	R/W
b26, b25	—	Reserved	These bits are read as 1. The write value should be 1.	R/W
b27	MSTPA27	Module Stop A27	Writing to and reading from this bit is enabled. When a transition to all-module clock stop mode is made, be sure that 1 has been written to this bit.	R/W
b28	MSTPA28	DMA Controller/Data Transfer Controller Module Stop	Target module: DMAC/DTC 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b29	MSTPA29	EXDMA Controller Module Stop	Target module: EXDMAC 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b30	—	Reserved	This bit is read as 1. The write value should be 1.	R/W
b31	ACSE	All-Module Clock Stop Mode Enable	0: All-module clock stop mode is disabled 1: All-module clock stop mode is enabled	R/W

ACSE Bit (All-Module Clock Stop Mode Enable)

The ACSE bit enables or disables a transition to all-module clock stop mode.

With the ACSE bit set to 1, when the CPU executes the WAIT instruction with the SBYCR.SSBY bit, MSTPCRA, MSTPCRB, MSTPCRC, and MSTPCRD satisfying specified conditions, the MCU enters all-module clock stop mode. For details, refer to section 11.6.2, All-Module Clock Stop Mode.

Whether to stop the 8-bit timers or not can be selected by the MSTPA5 and MSTPA4 bits.

When the MSTPCRA.ACSE bit = 0 while the SBYCR.SSBY bit = 0, a transition to sleep mode is made after the WAIT instruction is executed.

When the code flash memory P/E mode entry bit (FENTRYR.FENTRYC) in the flash P/E mode entry register is 1 or the data flash memory P/E mode entry bit (FENTRYR.FENTRYD) is 1, the setting of this bit is ineffective. Sleep mode is entered on execution of the WAIT instruction if this bit has been set to 1.

11.2.3 Module Stop Control Register B (MSTPCRB)

Address(es): 0008 0014h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	MSTPB 31	MSTPB 30	MSTPB 29	MSTPB 28	MSTPB 27	MSTPB 26	MSTPB 25	MSTPB 24	MSTPB 23	MSTPB 22	MSTPB 21	MSTPB 20	MSTPB 19	—	MSTPB 17	MSTPB 16
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	MSTPB 15	MSTPB 14	MSTPB 13	—	—	—	MSTPB 9	MSTPB 8	—	MSTPB 6	—	MSTPB 4	—	MSTPB 2	MSTPB 1	MSTPB 0
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Bit Name	Description	R/W
b0	MSTPB0	CAN Module 0 Module Stop*1	Target module: CAN0 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b1	MSTPB1	CAN Module 1 Module Stop*1	Target module: CAN1 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b2	MSTPB2	CAN Module 2 Module Stop*1	Target module: CAN2 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b3	—	Reserved	This bit is read as 1. The write value should be 1.	R/W
b4	MSTPB4	Serial Communication Interface SCIh Module Stop	Target module: SCIh (SCI12) 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b5	—	Reserved	This bit is read as 1. The write value should be 1.	R/W
b6	MSTPB6	Data Operation Circuit Module Stop	Target module: DOC 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b7	—	Reserved	This bit is read as 1. The write value should be 1.	R/W
b8	MSTPB8	Temperature Sensor Module Stop	Target module: Temperature sensor 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b9	MSTPB9	Event Link Controller Module Stop	Target module: ELC 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b12 to b10	—	Reserved	These bits are read as 1. The write value should be 1.	R/W
b13	MSTPB13	Ethernet-Controller PTP Controller and Ethernet-Controller DMA Controller Module Stop Setting*3, *4	Target modules: EPTPC, PTPEDMAC 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b14	MSTPB14	Ethernet Controller, Ethernet Controller DMA Controller, and PHY Management Interface (Channel 1) Modules Stop*4	Target modules: ETHERC, EDMAC, PMGI (channel 1) 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b15	MSTPB15	Ethernet Controller, Ethernet Controller DMA Controller, and PHY Management Interface (Channel 0) Modules Stop*4	Target modules: ETHERC, EDMAC, PMGI (channel 0) 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b16	MSTPB16	Serial Peripheral Interface 1 Module Stop	Target module: RSPI1 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b17	MSTPB17	Serial Peripheral Interface 0 Module Stop	Target module: RSPI0 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b18	—	Reserved	This bit is read as 1. The write value should be 1.	R/W

Bit	Symbol	Bit Name	Description	R/W
b19	MSTPB19	Universal Serial Bus 2.0 FS Interface Module Stop*2	Target module: USB0 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b20	MSTPB20	I2C Bus Interface 1 Module Stop	Target module: RIIC1 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b21	MSTPB21	I2C Bus Interface 0 Module Stop	Target module: RIIC0 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b22	MSTPB22	Parallel Data Capture Unit Module Stop	Target module: PDC 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b23	MSTPB23	CRC Calculator Module Stop	Target module: CRC 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b24	MSTPB24	Serial Communication Interface 7 Module Stop	Target module: SCI7 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b25	MSTPB25	Serial Communication Interface 6 Module Stop	Target module: SCI6 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b26	MSTPB26	Serial Communication Interface 5 Module Stop	Target module: SCI5 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b27	MSTPB27	Serial Communication Interface 4 Module Stop	Target module: SCI4 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b28	MSTPB28	Serial Communication Interface 3 Module Stop	Target module: SCI3 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b29	MSTPB29	Serial Communication Interface 2 Module Stop	Target module: SCI2 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b30	MSTPB30	Serial Communication Interface 1 Module Stop	Target module: SCI1 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b31	MSTPB31	Serial Communication Interface 0 Module Stop	Target module: SCI0 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W

Note 1. The MSTPBi bit should be rewritten while the oscillation of the clock controlled by the MSTPBi bit is stabilized. For entering software standby mode after writing a new value to the MSTPBi bit, wait for two cycles of the CAN clock (CANMCLK) to elapse after writing the new value, and then execute a WAIT instruction (i = 0 to 2).

Note 2. For entering software standby mode after writing a new value to the MSTPB19 bit, wait for two cycles of the USB clock (UCLK) to elapse after writing the new value, and then execute a WAIT instruction.

Note 3. Even if the EPTPC and PTPEDMAC modules are enabled (this MSTPB13 bit is set to 0), some registers of the EPTPC module may not be accessible. This depends on the combination of the settings of the MSTPB14, MSTPB15, and SYNFPn bypass (SYBYPSR.BYPASSn) (n = 0, 1) bits. For details, refer to section 37, PTP Module for the Ethernet Controller (EPTPCb).

Note 4. If any of the MSTPB13, MSTPB14, and MSTPB15 bits are set to release the corresponding modules from the module stop state, set the timeout detection enable bit (BEREN.TOEN) in the bus error monitoring enable register to 1.
For details of bus timeout, refer to section 16.7.1.2, Timeout.

11.2.4 Module Stop Control Register C (MSTPCRC)

Address(es): 0008 0018h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	MSTPC29	MSTPC28	MSTPC27	MSTPC26	MSTPC25	MSTPC24	MSTPC23	MSTPC22	—	—	MSTPC19	—	MSTPC17	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	MSTPC7	MSTPC6	—	—	—	MSTPC2	—	MSTPC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	MSTPC0	RAM Module Stop*1	Target module: RAM (0000 0000h to 0007 FFFFh) 0: RAM operating 1: RAM stopped	R/W
b1	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b2	MSTPC2	Expansion RAM Module Stop*2	Target module: Expansion RAM (0080 0000h to 0087 FFFFh) 0: Expansion RAM operating 1: Expansion RAM stopped	R/W
b5 to b3	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b6	MSTPC6	ECCRAM Module Stop*3	Target module: ECCRAM 0: ECCRAM operating 1: ECCRAM stopped	R/W
b7	MSTPC7	Standby RAM Module Stop*4	Target module: Standby RAM 0: Standby RAM operating 1: Standby RAM stopped	R/W
b15 to b8	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b16	—	Reserved	This bit is read as 1. The write value should be 1.	R/W
b17	MSTPC17	I2C Bus Interface 2 Module Stop	Target module: RIIC2 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b18	—	Reserved	This bit is read as 1. The write value should be 1.	R/W
b19	MSTPC19	CAC Module Stop*5	Target module: CAC 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b21, b20	—	Reserved	These bits are read as 1. The write value should be 1.	R/W
b22	MSTPC22	Serial Peripheral Interface 2 Module Stop	Target module: RSPI2 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b23	MSTPC23	Quad Serial Peripheral Interface Module Stop	Target module: QSPI 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b24	MSTPC24	Serial Communications Interface 11 Module Stop	Target module: SCI11 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b25	MSTPC25	Serial Communications Interface 10 Module Stop	Target module: SCI10 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b26	MSTPC26	Serial Communications Interface 9 Module Stop	Target module: SCI9 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b27	MSTPC27	Serial Communications Interface 8 Module Stop	Target module: SCI8 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W

Bit	Symbol	Bit Name	Description	R/W
b28	MSTPC28	2D drawing engine Module Stop	Target module: DRW2D 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b29	MSTPC29	Graphic-LCD controller Module Stop	Target module: GLCDC 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b31, b30	—	Reserved	These bits are read as 1. The write value should be 1.	R/W

Note 1. The MSTPC0 bit should not be set to 1 during access to the RAM. The RAM should not be accessed while the MSTPC0 bit is set to 1.

Note 2. The MSTPC2 bit should not be set to 1 during access to the expansion RAM. The expansion RAM should not be accessed while the MSTPC2 bit is set to 1.

Note 3. The MSTPC6 bit should not be set to 1 during access to the ECCRAM. The ECCRAM should not be accessed while the MSTPC6 bit is set to 1.

Note 4. The MSTPC7 bit should not be set to 1 during access to the standby RAM. The standby RAM should not be accessed while the MSTPC7 bit is set to 1.

Note 5. The MSTPC19 bit should be rewritten while the oscillation of the clock to be controlled by this bit is stable. For entering software standby mode after writing a new value to this bit, wait for two cycles of the slowest clock among the clocks output by the oscillators actually oscillating, and then execute a WAIT instruction.

11.2.5 Module Stop Control Register D (MSTPCRD)

Address(es): 0008 001Ch

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	MSTPD27	—	—	—	—	—	MSTPD21	—	MSTPD19	—	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	MSTPD15	MSTPD14	—	—	—	—	—	—	MSTPD7	MSTPD6	MSTPD5	MSTPD4	MSTPD3	MSTPD2	MSTPD1	MSTPD0
Value after reset:	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	MSTPD0	Module Stop D0	Writing to and reading from this bit is enabled. When a transition to all-module clock stop mode is made, be sure that 1 has been written to this bit.	R/W
b1	MSTPD1	Module Stop D1	Writing to and reading from this bit is enabled. When a transition to all-module clock stop mode is made, be sure that 1 has been written to this bit.	R/W
b2	MSTPD2	Module Stop D2	Writing to and reading from this bit is enabled. When a transition to all-module clock stop mode is made, be sure that 1 has been written to this bit.	R/W
b3	MSTPD3	Module Stop D3	Writing to and reading from this bit is enabled. When a transition to all-module clock stop mode is made, be sure that 1 has been written to this bit.	R/W
b4	MSTPD4	Module Stop D4	Writing to and reading from this bit is enabled. When a transition to all-module clock stop mode is made, be sure that 1 has been written to this bit.	R/W
b5	MSTPD5	Module Stop D5	Writing to and reading from this bit is enabled. When a transition to all-module clock stop mode is made, be sure that 1 has been written to this bit.	R/W
b6	MSTPD6	Module Stop D6	Writing to and reading from this bit is enabled. When a transition to all-module clock stop mode is made, be sure that 1 has been written to this bit.	R/W
b7	MSTPD7	Module Stop D7	Writing to and reading from this bit is enabled. When a transition to all-module clock stop mode is made, be sure that 1 has been written to this bit.	R/W
b13 to b8	—	Reserved	These bits are read as 1. The write value should be 1.	R/W
b14	MSTPD14	Extended Serial Sound Interface 1 Module Stop	Target module: SSIE1 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b15	MSTPD15	Extended Serial Sound Interface 0 Module Stop	Target module: SSIE0 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b18 to b16	—	Reserved	These bits are read as 1. The write value should be 1.	R/W
b19	MSTPD19	SD Host Interface Module Stop	Target module: SDHI 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b20	—	Reserved	This bit is read as 1. The write value should be 1.	R/W
b21	MSTPD21	MMC Host Interface Module Stop	Target module: MMCIF 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W
b26 to b22	—	Reserved	These bits are read as 1. The write value should be 1.	R/W
b27	MSTPD27	Trusted Secure IP Module Stop	Target module: Trusted Secure IP 0: Release from the module-stop state 1: Transition to the module-stop state is made	R/W

Bit	Symbol	Bit Name	Description	R/W
b31 to b28	—	Reserved	These bits are read as 1. The write value should be 1.	R/W

11.2.6 Operating Power Control Register (OPCCR)

Address(es): 0008 00A0h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	OPCM TSF	—	OPCM[2:0]		
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b2 to b0	OPCM[2:0]	Operating Power Control Mode Select	b2 b0 0 0 0: High-speed operating mode 1 1 0: Low-speed operating mode 1 1 1 1: Low-speed operating mode 2 Settings other than above are prohibited.	R/W
b3	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b4	OPCMTSF	Operating Power Control Mode Transition Status Flag	• Read 0: Transition completed 1: During transition • Write The write value should be 0.	R/W
b7 to b5	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The OPCCR register is used to reduce power consumption in normal operating mode, sleep mode, and all-module clock stop mode. Power consumption can be reduced according to the operating frequency and operating voltage to be used by the OPCCR setting.
There is no difference in power consumption between low speed mode 1 and 2 when the same conditions (frequency, voltage) are set in both modes.
OPCCR should not be modified in the following cases:

- When the operating power control mode transition status flag (OPCMTSF) is 1 (operating power control mode switching is in progress)
- When the code flash memory P/E mode entry bit (FENTRYR.FENTRYC) in the flash P/E mode entry register is 1 or the data flash memory P/E mode entry bit (FENTRYR.FENTRYD) is 1
- Period from the time of WAIT instruction execution for a sleep mode transition, to return from sleep mode to normal operation

For the procedure to use in shifting to operating power control mode, refer to section 11.5, Function for Lower Operating Power Consumption.
On return from software standby, the chip enters the high-speed operating mode. Even if a WAIT instruction is executed, if release from software standby precedes completion of the transition, the mode remains the same as before execution of the WAIT instruction. If this creates a problem, set the OPCCR.OPCM[2:0] bits to 000b during processing of the return interrupt.

OPCM[2:0] Bits (Operating Power Control Mode Select)

The OPCM[2:0] bits select operating power control mode in normal operating mode, sleep mode, and all-module clock stop mode.
Table 11.3 lists the operating power control modes along with the operating frequency ranges, operating voltage ranges, and power consumption.

Table 11.3 Relationship between Operating Power Control Mode, Operating Range, and Power Consumption

Operating Power Control Mode	OPCM [2:0] Bits	Operating Frequency Range							Operating Voltage Range		Power Consumption
		Reading Flash Memory						Programming or Erasing Flash Memory	Reading Flash Memory	Programming or Erasing Flash Memory	
		ICLK	FCLK	PCLKA	PCLKB	PCLKC PCLKD	BCLK	FCLK			
High-speed operating mode	000b	240 MHz max	60 MHz max	120 MHz max	60 MHz max	60 MHz max*1	120 MHz max	4 MHz to 60 MHz	2.7 V to 3.6 V	2.7 V to 3.6 V	High
Low-speed operating mode 1*2	110b	1 MHz max	1 MHz max	1 MHz max	1 MHz max	1 MHz max*1	1 MHz max	P/E disabled	2.7 V to 3.6 V	P/E disabled	↓
Low-speed operating mode 2*2	111b	32 kHz to 264 kHz	32 kHz to 264 kHz	264 kHz max	264 kHz max	264 kHz max*1	264 kHz max	P/E disabled	2.7 V to 3.6 V	P/E disabled	Low

Note 1. When the 12-bit A/D converter is used in high-speed operating mode or low-speed operating mode 1, the frequency must be set to at least 1 MHz. The 12-bit A/D converter cannot be used in low-speed operating mode 2.

Note 2. There is no difference in power consumption between low speed mode 1 and 2 when the same conditions (frequency, voltage) are set in both modes.

Each operating power control mode is described below.

- High-speed operating mode

This mode allows high-speed operation.
For reading the flash memory (FLASH), the maximum operating frequency of ICLK is 240 MHz, that of PCLKA and BCLK are 120 MHz, and that of FCLK and PCLKB are 60 MHz. The S12AD conversion clocks PCLKC and PCLKD can be operated in the operating frequency from 1 MHz to 60 MHz. During Flash memory programming/erasure (P/E), the FCLK can be operated in the operating frequency from 4 MHz to 60 MHz. When the frequency of ICLK is set to faster than 120 MHz, the value of the MEMWAIT register needs to be modified. The operating voltage is in the range of 2.7 to 3.6 V both for Flash memory read and P/E. After release from a reset, this MCU is activated in this mode.

- Low-speed operating mode 1

This mode reduces power consumption for low-speed operation.
During reading the flash memory (FLASH), the maximum operating frequency of ICLK, FCLK, PCLKA, PCLKB and BCLK is 1 MHz. The operating voltage is in the range of 2.7 to 3.6 V.
In low-speed operating mode 1, P/E operation of flash memory is disabled. Writing to set the PLLCR2.PLLEN (PLL operation) or PPLLCR2.PPLLEN bit to 0 (PPLL operation) is prohibited.
In this mode, lower power consumption is possible than in high-speed operating mode when the same operation is performed under the same conditions (operating frequency, operating voltage).

- Low-speed operating mode 2

As compare to low-speed operating mode 1, this mode reduces power consumption for low-speed operation.
During reading the flash memory (FLASH), the maximum operating frequency of ICLK, FCLK, PCLKA, PCLKB and BCLK is 264 kHz, and the minimum operating frequency of ICLK and FCLK is 32 kHz. The operating voltage is in the range of 2.7 to 3.6 V.
The following restrictions apply when low-speed operating mode 2 is selected:

- P/E operations for flash memory are prohibited.
- Using the PLL, HOCO or PPLL is prohibited.
- Using the oscillation stop detection function of the main clock oscillator is prohibited.

When the clock source select bits in the system clock control register 3 (SCKCR3.CKSEL[2:0]) are 011b (sub-clock oscillator selected) and the system clock select bits (ICK[3:0]) or Flash-IF clock select bits (FCK[3:0]) in the system clock control register (SCKCR) are not 0000b (no frequency dividing), OPCM[2:0] bits cannot be set to 111b.
When the PLL stop control bit (PLLCR2.PLLEN) in PLL control register 2 is 0 (PLL operation), writing 110b (low-speed operating mode 1) and 111b (low-speed operating mode 2) to the OPCM[2:0] bits is not possible.
In addition, when the PPLL stop control bit (PPLLCR2.PPLLEN) in the PPLL control register 2 is 0 (PPLL operation), writing 110b (low-speed operating mode 1) and 111b (low-speed operating mode 2) to the OPCM[2:0] bits is not possible.

OPCMTSF Flag (Operating Power Control Mode Transition Status Flag)

The OPCMTSF flag indicates the switching control state when the operating power control mode is switched.
When a write access is attempted to change the operating power control mode, the OPCMTSF flag is set to 1. The flag becomes 0 after a transition to the changed control mode is completed. Make sure that the OPCMTSF flag is 0 (completed operating power control mode transition) before the next processing.

11.2.7 Sleep Mode Return Clock Source Switching Register (RSTCKCR)

Address(es): 0008 00A1h

	b7	b6	b5	b4	b3	b2	b1	b0
	RSTCKEN	—	—	—	—	RSTCKSEL[2:0]		
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b2 to b0	RSTCKSEL [2:0]	Sleep Mode Return Clock Source Select	b2 b0 0 0 1: HOCO is selected 0 1 0: Main clock oscillator is selected Settings other than above are prohibited while the RSTCKEN bit is 1.	R/W
b6 to b3	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b7	RSTCKEN	Sleep Mode Return Clock Source Switching Enable	0: Clock source switching on release from sleep mode is disabled 1: Clock source switching on release from sleep mode is enabled	R/W

The RSTCKCR register is used to control clock source switching at the time of release from sleep mode.
When operation is restored from sleep mode by setting RSTCKCR, the main clock oscillator stop bit in the main clock oscillator control register (MOSCCR.MOSTP) and HOCO stop bit in the high-speed on-chip oscillator control register (HOCOCR.HCSTP) corresponding to the clock source to be used on restoration are automatically modified to the operating state. The value of RSTCKSEL[2:0] bits is automatically reloaded to the clock source select bits in the system clock control register 3 (SCKCR3.CKSEL[2:0]).
The sleep mode return clock source switching function and clock source switching function by the ELC cannot be used at the same time. To enable the sleep mode return clock source switching function, write 1 to the RSTCKCR.RSTCKEN bit with the ELC clock source switching function disabled. The ELC clock source switching function should be enabled with the RSTCKCR.RSTCKEN bit being 0.
When the setting of register RSTCKCR is for the HOCO to be used in recovery from sleep mode, the power supply for the HOCO is not automatically switched on. If the HOCO to be used in recovery from sleep mode, the power supply for the HOCO must be on when the transition to sleep mode takes place.
When return from sleep mode is made while clock source switching on release from sleep mode is enabled (RSTCKCR.RSTCKEN is 1), and operating power control mode select bits (OPCCR.OPCM[2:0]) are set so as to select low-speed operating mode 1 (110b) or low-speed operating mode 2 (111b), the OPCCR.OPCM[2:0] bits are automatically switched to high-speed mode (000b).

RSTCKSEL[2:0] Bits (Sleep Mode Return Clock Source Select)

The RSTCKSEL[2:0] bits select the clock source to be used at the time of release from sleep mode.
The clock source selected by the RSTCKSEL[2:0] bits is enabled only when the RSTCKEN bit is 1.

RSTCKEN Bit (Sleep Mode Return Clock Source Switching Enable)

The RSTCKEN bit enables or disables clock source switching at the time of release from sleep mode.
On release from sleep mode, the clock source should be switched only when LOCO or sub clock is selected as a clock for a transition to sleep mode. To make a transition to sleep mode with HOCO, main clock, or PLL selected as the clock source, the RSTCKEN bit should not be set to 1.

11.2.8 Deep Standby Control Register (DPSBYCR)

Address(es): 0008 C280h

	b7	b6	b5	b4	b3	b2	b1	b0
	DPSBY	IOKEEP	—	—	—	—	DEEPCUT[1:0]	
Value after reset:	0	0	0	0	0	0	0	1

Bit	Symbol	Bit Name	Description	R/W
b1, b0	DEEPCUT [1:0]	Deep Cut	b1 b0 0 0: Power is supplied to the standby RAM and USB resume detecting unit in deep software standby mode 0 1: Power is not supplied to the standby RAM and USB resume detecting unit in deep software standby mode 1 0: Setting prohibited 1 1: Power is not supplied to the standby RAM and USB resume detecting unit in deep software standby mode. In addition, LVD is stopped and the low power consumption function in a power-on reset circuit is enabled.	R/W
b5 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b6	IOKEEP	I/O Port Retention	0: Release from deep software standby mode and cancellation of I/O port retention proceed simultaneously. 1: The I/O port state is retained even after release from deep software standby mode. Then, writing 0 to the IOKEEP bit cancels the I/O port retention.	R/W
b7	DPSBY	Deep Software Standby	SSBY b7 0 0: Transition to sleep mode or all-module clock stop mode is made after the WAIT instruction is executed 0 1: Transition to sleep mode or all-module clock stop mode is made after the WAIT instruction is executed 1 0: Transition to software standby mode is made after the WAIT instruction is executed 1 1: Transition to deep software standby mode is made after the WAIT instruction is executed	R/W

The DPSBYCR register is not initialized by the internal reset signal that is the source for release from deep software standby mode. For details, refer to Table 6.2, Targets to be Initialized by Each Reset Source.

DEEPCUT[1:0] Bits (Deep Cut)

The DEEPCUT[1:0] bits control the internal power supply to the standby RAM and USB resume detecting unit in deep software standby mode. In addition, these bits control the state of LVD and power-on reset circuit in deep software standby mode.

The internal power supply of standby RAM and USB resume detecting unit can be controlled by the setting of the DEEPCUT[1:0] bits.

When a USB suspend/resume interrupt*1 is used as a deep software standby mode releasing source, the DEEPCUT[1:0] bits must be set to 00b.

When the LVD is used in deep software standby mode, the DEEPCUT[1:0] bits must be set to 00b or 01b.

For lower power consumption, set the DEEPCUT[1:0] bits to 11b so that the LVD is stopped and the low power consumption function of the power-on reset circuit is enabled.

Note 1. USB is not released from deep software standby mode using USB0_OVRCURB multiplexed with pin P22.

IOKEEP Bit (I/O Port Retention)

In deep software standby mode, I/O ports keep retaining the same states from software standby mode. The IOKEEP bit specifies whether to keep retaining the I/O port states from deep software standby mode even after release from deep software standby mode, or to cancel retention of the I/O port states.

DPSBY Bit (Deep Software Standby)

The DPSBY bit controls transitions to deep software standby mode.
When the WAIT instruction is executed while the SBYCR.SSBY and DPSBY bits are both 1, the MCU enters deep software standby mode through software standby mode.
The DPSBY bit remains 1 when release from deep software standby mode is triggered by certain pins which are sources of external pin interrupts (NMI, IRQ0-DS to IRQ15-DS, SCL2-DS, SDA2-DS, and CRX1-DS) or a peripheral interrupt (RTC alarm, RTC periodic, USB suspend/resume*1, voltage monitoring 1, or voltage monitoring 2). Write 0 to this bit to clear it.
The setting of the DPSBY bit becomes invalid when the IWDT is in auto-start mode and the OFS0.IWDTSLCSTP is 0 (counting continues) or the IWDT is in register start mode and the SLCSTP bit in IWDTCSTPR is 0.
Instead, even when the SBYCR.SSBY bit is 1 and the DPSBY bit 1, the transition after the execution of a WAIT instruction is to software standby mode.
The setting of the DPSBY bit becomes invalid when voltage monitoring 1 reset is enabled by the voltage monitoring 1 circuit mode select bit (LVD1CR0.LVD1RI = 1) or when a voltage monitoring 2 reset is enabled by the voltage monitoring 2 circuit mode bit (LVD2CR0.LVD2RI = 1). In this case, even when the SBYCR.SSBY bit is 1 and the DPSBY bit is 1, the transition after the execution of a WAIT instruction is to software standby mode.

Note 1. USB is not released from deep software standby mode using USB0_OVRCURB multiplexed with pin P22.

11.2.9 Deep Standby Interrupt Enable Register 0 (DPSIER0)

Address(es): 0008 C282h

	b7	b6	b5	b4	b3	b2	b1	b0
	DIRQ7E	DIRQ6E	DIRQ5E	DIRQ4E	DIRQ3E	DIRQ2E	DIRQ1E	DIRQ0E
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	DIRQ0E	IRQ0-DS Pin Enable	0: Release from deep software standby mode by the IRQ0-DS pin is disabled 1: Release from deep software standby mode by the IRQ0-DS pin is enabled	R/W
b1	DIRQ1E	IRQ1-DS Pin Enable	0: Release from deep software standby mode by the IRQ1-DS pin is disabled 1: Release from deep software standby mode by the IRQ1-DS pin is enabled	R/W
b2	DIRQ2E	IRQ2-DS Pin Enable	0: Release from deep software standby mode by the IRQ2-DS pin is disabled 1: Release from deep software standby mode by the IRQ2-DS pin is enabled	R/W
b3	DIRQ3E	IRQ3-DS Pin Enable	0: Release from deep software standby mode by the IRQ3-DS pin is disabled 1: Release from deep software standby mode by the IRQ3-DS pin is enabled	R/W
b4	DIRQ4E	IRQ4-DS Pin Enable	0: Release from deep software standby mode by the IRQ4-DS pin is disabled 1: Release from deep software standby mode by the IRQ4-DS pin is enabled	R/W
b5	DIRQ5E	IRQ5-DS Pin Enable	0: Release from deep software standby mode by the IRQ5-DS pin is disabled 1: Release from deep software standby mode by the IRQ5-DS pin is enabled	R/W
b6	DIRQ6E	IRQ6-DS Pin Enable	0: Release from deep software standby mode by the IRQ6-DS pin is disabled 1: Release from deep software standby mode by the IRQ6-DS pin is enabled	R/W
b7	DIRQ7E	IRQ7-DS Pin Enable	0: Release from deep software standby mode by the IRQ7-DS pin is disabled 1: Release from deep software standby mode by the IRQ7-DS pin is enabled	R/W

The DPSIER0 register is not initialized by the internal reset signal used as deep software standby mode releasing source. For details, refer to Table 6.2, Targets to be Initialized by Each Reset Source.

After the setting of DPSIER0 is modified, an edge may be internally generated depending on the state of the pin, resulting in DPSIFR0 being set to 1. Therefore, DPSIFR0 should be set to 0 before a transition to deep software standby mode.

11.2.10 Deep Standby Interrupt Enable Register 1 (DPSIER1)

Address(es): 0008 C283h

	b7	b6	b5	b4	b3	b2	b1	b0
	DIRQ15E	DIRQ14E	DIRQ13E	DIRQ12E	DIRQ11E	DIRQ10E	DIRQ9E	DIRQ8E
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	DIRQ8E	IRQ8-DS Pin Enable	0: Release from deep software standby mode by the IRQ8-DS pin is disabled 1: Release from deep software standby mode by the IRQ8-DS pin is enabled	R/W
b1	DIRQ9E	IRQ9-DS Pin Enable	0: Release from deep software standby mode by the IRQ9-DS pin is disabled 1: Release from deep software standby mode by the IRQ9-DS pin is enabled	R/W
b2	DIRQ10E	IRQ10-DS Pin Enable	0: Release from deep software standby mode by the IRQ10-DS pin is disabled 1: Release from deep software standby mode by the IRQ10-DS pin is enabled	R/W
b3	DIRQ11E	IRQ11-DS Pin Enable	0: Release from deep software standby mode by the IRQ11-DS pin is disabled 1: Release from deep software standby mode by the IRQ11-DS pin is enabled	R/W
b4	DIRQ12E	IRQ12-DS Pin Enable	0: Release from deep software standby mode by the IRQ12-DS pin is disabled 1: Release from deep software standby mode by the IRQ12-DS pin is enabled	R/W
b5	DIRQ13E	IRQ13-DS Pin Enable	0: Release from deep software standby mode by the IRQ13-DS pin is disabled 1: Release from deep software standby mode by the IRQ13-DS pin is enabled	R/W
b6	DIRQ14E	IRQ14-DS Pin Enable	0: Release from deep software standby mode by the IRQ14-DS pin is disabled 1: Release from deep software standby mode by the IRQ14-DS pin is enabled	R/W
b7	DIRQ15E	IRQ15-DS Pin Enable	0: Release from deep software standby mode by the IRQ15-DS pin is disabled 1: Release from deep software standby mode by the IRQ15-DS pin is enabled	R/W

The DPSIER1 register is not initialized by the internal reset signal used as deep software standby mode releasing source. For details, refer to Table 6.2, Targets to be Initialized by Each Reset Source.

After the setting of DPSIER1 is modified, an edge may be internally generated depending on the state of the pin, resulting in DPSIFR1 being set to 1. Therefore, DPSIFR1 should be set to 0 before a transition to deep software standby mode.

11.2.11 Deep Standby Interrupt Enable Register 2 (DPSIER2)

Address(es): 0008 C284h

	b7	b6	b5	b4	b3	b2	b1	b0
	DUSBIE	DRIICCIE	DRIICDIE	DNMIE	DRTCAIE	DRTCIIE	DLVD2IE	DLVD1IE
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	DLVD1IE	LVD1 Deep Standby Release Signal Enable	0: Disable release from deep software standby mode by the voltage monitoring 1 signal 1: Enable release from deep software standby mode by the voltage monitoring 1 signal	R/W
b1	DLVD2IE	LVD2 Deep Standby Release Signal Enable	0: Disable release from deep software standby mode by the voltage monitoring 2 signal 1: Enable release from deep software standby mode by the voltage monitoring 2 signal	R/W
b2	DRTCIIE	RTC Periodic Interrupt Deep Standby Release Signal Enable	0: Release from deep software standby mode by the RTC periodic interrupt signal is disabled 1: Release from deep software standby mode by the RTC periodic interrupt signal is enabled	R/W
b3	DRTCAIE	RTC Alarm Interrupt Deep Standby Release Signal Enable	0: Release from deep software standby mode by the RTC alarm interrupt signal is disabled 1: Release from deep software standby mode by the RTC alarm interrupt signal is enabled	R/W
b4	DNMIE	NMI Pin Enable	0: Release from deep software standby mode by the NMI pin is disabled 1: Release from deep software standby mode by the NMI pin is enabled	R/W*1
b5	DRIICDIE	SDA2-DS Deep Standby Release Signal Enable	0: Release from deep software standby mode by the SDA2-DS signal is disabled 1: Release from deep software standby mode by the SDA2-DS signal is enabled	R/W
b6	DRIICCIE	SCL2-DS Deep Standby Release Signal Enable	0: Release from deep software standby mode by the SCL2-DS signal is disabled 1: Release from deep software standby mode by the SCL2-DS signal is enabled	R/W
b7	DUSBIE	USB Suspend/Resume Deep Standby Release Signal Enable	0: Release from deep software standby mode by the USB suspend/resume is disabled 1: Release from deep software standby mode by the USB suspend/resume is enabled	R/W

Note 1. 1 can be written only once. Once 1 is written to this bit, subsequent write accesses are disabled.

The DPSIER2 register is not initialized by the internal reset signal used as deep software standby mode releasing source. For details, refer to Table 6.2, Targets to be Initialized by Each Reset Source.
After the setting of DPSIER2 is modified, an edge may be internally generated depending on the state of the pin, resulting in DPSIFR2 being set to 1. Therefore, DPSIFR2 should be set to 0 before a transition to deep software standby mode.

DUSBIE Bit (USB Suspend/Resume Deep Standby Release Signal Enable)

The DUSBIE bit is an enable bit for USB. USB is not released from deep software standby mode using USB0_OVRCURB multiplexed with pin P22.

11.2.12 Deep Standby Interrupt Enable Register 3 (DPSIER3)

Address(es): 0008 C285h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	DCANIE
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	DCANIE	CRX1-DS Deep Standby Release Signal Enable	0: Release from deep software standby mode by the CRX1-DS pin is disabled 1: Release from deep software standby mode by the CRX1-DS pin is enabled	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The DPSIER3 register is not initialized by the internal reset signal used as deep software standby mode releasing source. For details, refer to Table 6.2, Targets to be Initialized by Each Reset Source.

After the setting of DPSIER3 is modified, an edge may be internally generated depending on the state of the pin, resulting in DPSIFR3 being set to 1. Therefore, DPSIFR3 should be set to 0 before a transition to deep software standby mode.

11.2.13 Deep Standby Interrupt Flag Register 0 (DPSIFR0)

Address(es): 0008 C286h

	b7	b6	b5	b4	b3	b2	b1	b0
	DIRQ7F	DIRQ6F	DIRQ5F	DIRQ4F	DIRQ3F	DIRQ2F	DIRQ1F	DIRQ0F
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	DIRQ0F	IRQ0-DS Pin Deep Standby Release Flag	0: Request for release is not being generated on the IRQ0-DS pin 1: Request for release is being generated on the IRQ0-DS pin	R(/W) *1
b1	DIRQ1F	IRQ1-DS Pin Deep Standby Release Flag	0: Request for release is not being generated on the IRQ1-DS pin 1: Request for release is being generated on the IRQ1-DS pin	R(/W) *1
b2	DIRQ2F	IRQ2-DS Pin Deep Standby Release Flag	0: Request for release is not being generated on the IRQ2-DS pin 1: Request for release is being generated on the IRQ2-DS pin	R(/W) *1
b3	DIRQ3F	IRQ3-DS Pin Deep Standby Release Flag	0: Request for release is not being generated on the IRQ3-DS pin 1: Request for release is being generated on the IRQ3-DS pin	R(/W) *1
b4	DIRQ4F	IRQ4-DS Pin Deep Standby Release Flag	0: Request for release is not being generated on the IRQ4-DS pin 1: Request for release is being generated on the IRQ4-DS pin	R(/W) *1
b5	DIRQ5F	IRQ5-DS Pin Deep Standby Release Flag	0: Request for release is not being generated on the IRQ5-DS pin 1: Request for release is being generated on the IRQ5-DS pin	R(/W) *1
b6	DIRQ6F	IRQ6-DS Pin Deep Standby Release Flag	0: Request for release is not being generated on the IRQ6-DS pin 1: Request for release is being generated on the IRQ6-DS pin	R(/W) *1
b7	DIRQ7F	IRQ7-DS Pin Deep Standby Release Flag	0: Request for release is not being generated on the IRQ7-DS pin 1: Request for release is being generated on the IRQ7-DS pin	R(/W) *1

Note 1. Only 0 can be written to clear the flag.

Each flag is set to 1 when a request for release specified by the DPSIEGR0 register is generated.
Each flag may be set to 1 when a request for release is generated in any mode (not even deep software standby mode) or when the setting of the DPSIER0 register is modified. Therefore, a transition to deep software standby mode should be made after the DPSIFR0 register is set to 00h.
To set the DPSIFR0 register to 00h after modifying the DPSIER0 register, wait for at least six PCLKB cycles, read the DPSIFR0 register, and then write 0 to the DPSIFR0 register. Six or more PCLKB cycles can be secured, for example, by reading the DPSIER0 register.
The DPSIFR0 register is not initialized by the internal reset signal used as deep software standby mode releasing source. For details, refer to Table 6.2, Targets to be Initialized by Each Reset Source.

DIRQnF Flags (IRQn Deep Standby Release Flag) (n = 0 to 7)

These flags indicate that a request for release has been generated on the IRQn-DS pin.
[Setting condition]

- A request for release is generated on the IRQn-DS pin specified by the DPSIEGR0 register

[Clearing condition]

- Each bit is read as 1 and then written by 0

11.2.14 Deep Standby Interrupt Flag Register 1 (DPSIFR1)

Address(es): 0008 C287h

	b7	b6	b5	b4	b3	b2	b1	b0
	DIRQ15F	DIRQ14F	DIRQ13F	DIRQ12F	DIRQ11F	DIRQ10F	DIRQ9F	DIRQ8F
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	DIRQ8F	IRQ8-DS Pin Deep Standby Release Flag	0: Request for release is not being generated on the IRQ8-DS pin 1: Request for release is being generated on the IRQ8-DS pin	R(/W) *1
b1	DIRQ9F	IRQ9-DS Pin Deep Standby Release Flag	0: Request for release is not being generated on the IRQ9-DS pin 1: Request for release is being generated on the IRQ9-DS pin	R(/W) *1
b2	DIRQ10F	IRQ10-DS Pin Deep Standby Release Flag	0: Request for release is not being generated on the IRQ10-DS pin 1: Request for release is being generated on the IRQ10-DS pin	R(/W) *1
b3	DIRQ11F	IRQ11-DS Pin Deep Standby Release Flag	0: Request for release is not being generated on the IRQ11-DS pin 1: Request for release is being generated on the IRQ11-DS pin	R(/W) *1
b4	DIRQ12F	IRQ12-DS Pin Deep Standby Release Flag	0: Request for release is not being generated on the IRQ12-DS pin 1: Request for release is being generated on the IRQ12-DS pin	R(/W) *1
b5	DIRQ13F	IRQ13-DS Pin Deep Standby Release Flag	0: Request for release is not being generated on the IRQ13-DS pin 1: Request for release is being generated on the IRQ13-DS pin	R(/W) *1
b6	DIRQ14F	IRQ14-DS Pin Deep Standby Release Flag	0: Request for release is not being generated on the IRQ14-DS pin 1: Request for release is being generated on the IRQ14-DS pin	R(/W) *1
b7	DIRQ15F	IRQ15-DS Pin Deep Standby Release Flag	0: Request for release is not being generated on the IRQ15-DS pin 1: Request for release is being generated on the IRQ15-DS pin	R(/W) *1

Note 1. Only 0 can be written to clear the flag.

Each flag is set to 1 when a request for release specified by the DPSIEGR1 register is generated.
Each flag may be set to 1 when a request for release is generated in any mode (not even deep software standby mode) or when the setting of the DPSIER1 register is modified. Therefore, a transition to deep software standby mode should be made after the DPSIFR1 register is set to 00h.
To set the DPSIFR1 register to 00h after modifying the DPSIER1 register, wait for at least six PCLKB cycles, read the DPSIFR1 register, and then write 0 to the DPSIFR1 register. Six or more PCLKB cycles can be secured, for example, by reading the DPSIER1 register.
The DPSIFR1 register is not initialized by the internal reset signal used as deep software standby mode releasing source. For details, refer to Table 6.2, Targets to be Initialized by Each Reset Source.

DIRQnF Flags (IRQn Deep Standby Release Flag) (n = 8 to 15)

These flags indicate that a request for release has been generated on the IRQn-DS pin.
[Setting condition]

- A request for release is generated on the IRQn-DS pin specified by the DPSIEGR1 register

[Clearing condition]

- Each bit is read as 1 and then written by 0

11.2.15 Deep Standby Interrupt Flag Register 2 (DPSIFR2)

Address(es): 0008 C288h

	b7	b6	b5	b4	b3	b2	b1	b0
	DUSBIF	DRIICCIF	DRIICDIF	DNMIF	DRTCAIF	DRTCIIF	DLVD2IF	DLVD1IF
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	DLVD1IF	LVD1 Deep Standby Release Flag	0: Request for release by the voltage monitor 1 signal is not being generated 1: Request for release by the voltage monitor 1 signal is being generated	R(/W) *1
b1	DLVD2IF	LVD2 Deep Standby Release Flag	0: Request for release by the voltage monitor 2 signal is not being generated 1: Request for release by the voltage monitor 2 signal is being generated	R(/W) *1
b2	DRTCIIF	RTC Periodic Interrupt Deep Standby Release Flag	0: Request for release by the RTC periodic interrupt signal is not being generated 1: Request for release by the RTC periodic interrupt signal is being generated	R(/W) *1
b3	DRTCAIF	RTC Alarm Interrupt Deep Standby Release Flag	0: Request for release by the RTC alarm interrupt signal is not being generated 1: Request for release by the RTC alarm interrupt signal is being generated	R(/W) *1
b4	DNMIF	NMI Deep Standby Release Flag	0: Request for release is not being generated on the NMI pin 1: Request for release is being generated on the NMI pin	R(/W) *1
b5	DRIICDIF	SDA2-DS Deep Standby Release Flag	0: Request for release by the SDA2-DS signal is not being generated 1: Request for release by the SDA2-DS signal is being generated	R(/W) *1
b6	DRIICCIF	SCL2-DS Deep Standby Release Flag	0: Request for release by the SCL2-DS signal is not being generated 1: Request for release by the SCL2-DS signal is being generated	R(/W) *1
b7	DUSBIF	USB Suspend/Resume Deep Standby Release Flag	0: Request for release by the USB suspend/resume is not being generated 1: Request for release by the USB suspend/resume is being generated	R(/W) *1

Note 1. Only 0 can be written to clear the flag.

Each flag is set to 1 when a request for release specified by the DPSIEGR2 register is generated.
Each flag may be set to 1 when a request for release is generated in any mode (not even deep software standby mode) or when the setting of the DPSIER2 register is modified. Therefore, a transition to deep software standby mode should be made after the DPSIFR2 register is set to 00h.
To set the DPSIFR2 register to 00h after modifying the DPSIER2 register, wait for at least six PCLKB cycles, read the DPSIFR2 register, and then write 0 to the DPSIFR2 register. Six or more PCLKB cycles can be secured, for example, by reading the DPSIER2 register.
The DPSIFR2 register is not initialized by the internal reset signal used as deep software standby mode releasing source. For details, refer to Table 6.2, Targets to be Initialized by Each Reset Source.

DLVDmIF Flag (LVDm Deep Standby Release Flag) (m = 1, 2)

This flag indicates that a request for release by the voltage monitor m signal has been generated.
[Setting condition]

- A request for release is generated by the voltage monitoring m signal that is selected in the DPSIEGR2 register

[Clearing condition]

- This bit is read as 1 and then written by 0

DRTCIIF Flag (RTC Periodic Interrupt Deep Standby Release Flag)

This flag indicates that a request for release by the RTC periodic interrupt signal has been generated.
[Setting condition]

- A request for release by the RTC periodic interrupt signal is generated

[Clearing condition]

- This bit is read as 1 and then written by 0

DRTCAIF Flag (RTC Alarm Interrupt Deep Standby Release Flag)

This flag indicates that a request for release by the RTC alarm interrupt signal has been generated.
[Setting condition]

- A request for release by the RTC alarm interrupt signal is generated

[Clearing condition]

- This bit is read as 1 and then written by 0

DNMIF Flag (NMI Deep Standby Release Flag)

This flag indicates that a request for release has been generated on the NMI pin.
[Setting condition]

- A request for release is generated on the NMI pin specified by DPSIEGR2

[Clearing condition]

- This bit is read as 1 and then written by 0

DRIICDIF Flag (SDA2-DS Deep Standby Release Flag)

This flag indicates that a request for release by the SDA2-DS interrupt signal has been generated.
[Setting condition]

- A request for release is generated on the SDA2-DS pin specified by DPSIEGR2

[Clearing condition]

- This bit is read as 1 and then written by 0

DRIICCIF Flag (SCL2-DS Deep Standby Release Flag)

This flag indicates that a request for release by the SCL2-DS interrupt signal has been generated.
[Setting condition]

- A request for release is generated on by the SCL2-DS pin specified by DPSIEGR2

[Clearing condition]

- This bit is read as 1 and then written by 0

DUSBIF Flag (USB Suspend/Resume Deep Standby Release Flag)

This flag indicates that a request for release by the USB suspend/resume has been generated.
The DUSBIF flag is a flag for USB. However, USB is not released from deep software standby mode using USB0_OVRCURB multiplexed with pin P22.
Figure 11.2 shows the configuration of the DUSBIF flag.
[Setting condition]

- A request for release by the USB suspend/resume is generated

[Clearing condition]

- This bit is read as 1 and then written by 0

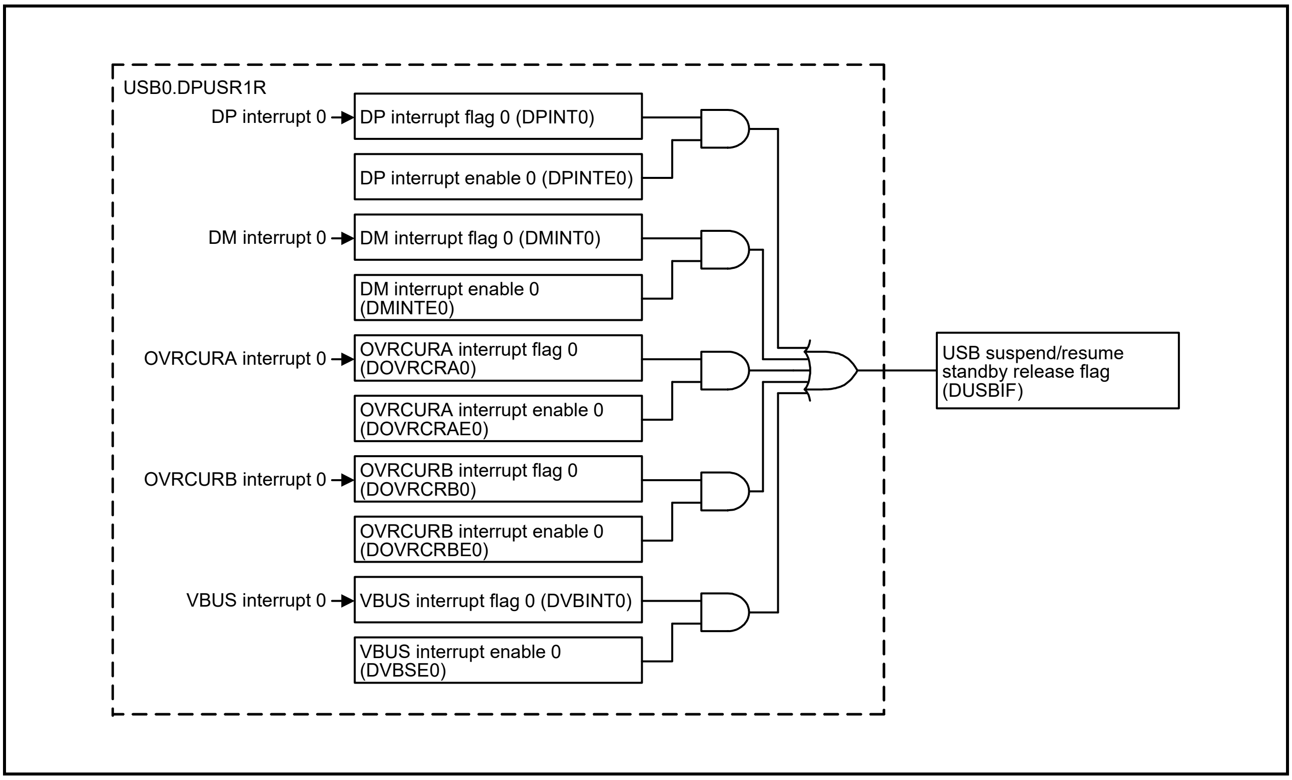


Figure 11.2 Configuration of the USB Suspend/Resume Deep Standby Release Flag (DUSBIF)

11.2.16 Deep Standby Interrupt Flag Register 3 (DPSIFR3)

Address(es): 0008 C289h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	DCANIF
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	DCANIF	CRX1-DS Deep Standby Release Flag	0: Request for release is not being generated on the CRX1-DS pin 1: Request for release is being generated on the CRX1-DS pin	R(/W) *1
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Only 0 can be written to clear the flag.

Each flag is set to 1 when a request for release specified by the DPSIEGR3 register is generated.
Each flag may be set to 1 when a request for release is generated in any mode (not even deep software standby mode) or when the setting of the DPSIER3 register is modified. Therefore, a transition to deep software standby mode should be made after the DPSIFR3 register is set to 00h.
To set the DPSIFR3 register to 00h after modifying the DPSIER3 register, wait for at least six PCLKB cycles, read the DPSIFR3 register, and then write 0 to the DPSIFR3 register. Six or more PCLKB cycles can be secured, for example, by reading the DPSIER3 register.
The DPSIFR3 register is not initialized by the internal reset signal used as deep software standby mode releasing source. For details, refer to Table 6.2, Targets to be Initialized by Each Reset Source.

DCANIF Flag (CRX1-DS Deep Standby Release Flag)

This flag indicates that a request for release has been generated on the CRX1-DS pin.
[Setting condition]

- A request for release is generated on the CRX1-DS pin specified by the DPSIEGR3 register

[Clearing condition]

- This bit is read as 1 and then written by 0

11.2.17 Deep Standby Interrupt Edge Register 0 (DPSIEGR0)

Address(es): 0008 C28Ah

	b7	b6	b5	b4	b3	b2	b1	b0
	DIRQ7EG	DIRQ6EG	DIRQ5EG	DIRQ4EG	DIRQ3EG	DIRQ2EG	DIRQ1EG	DIRQ0EG
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	DIRQ0EG	IRQ0-DS Pin Edge Select	0: A request for release is generated at a falling edge 1: A request for release is generated at a rising edge	R/W
b1	DIRQ1EG	IRQ1-DS Pin Edge Select	0: A request for release is generated at a falling edge 1: A request for release is generated at a rising edge	R/W
b2	DIRQ2EG	IRQ2-DS Pin Edge Select	0: A request for release is generated at a falling edge 1: A request for release is generated at a rising edge	R/W
b3	DIRQ3EG	IRQ3-DS Pin Edge Select	0: A request for release is generated at a falling edge 1: A request for release is generated at a rising edge	R/W
b4	DIRQ4EG	IRQ4-DS Pin Edge Select	0: A request for release is generated at a falling edge 1: A request for release is generated at a rising edge	R/W
b5	DIRQ5EG	IRQ5-DS Pin Edge Select	0: A request for release is generated at a falling edge 1: A request for release is generated at a rising edge	R/W
b6	DIRQ6EG	IRQ6-DS Pin Edge Select	0: A request for release is generated at a falling edge 1: A request for release is generated at a rising edge	R/W
b7	DIRQ7EG	IRQ7-DS Pin Edge Select	0: A request for release is generated at a falling edge 1: A request for release is generated at a rising edge	R/W

The DPSIEGR0 register is not initialized by the internal reset signal that is the source for release from deep software standby mode. For details, refer to Table 6.2, Targets to be Initialized by Each Reset Source.

11.2.18 Deep Standby Interrupt Edge Register 1 (DPSIEGR1)

Address(es): 0008 C28Bh

	b7	b6	b5	b4	b3	b2	b1	b0
	DIRQ15EG	DIRQ14EG	DIRQ13EG	DIRQ12EG	DIRQ11EG	DIRQ10EG	DIRQ9EG	DIRQ8EG
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	DIRQ8EG	IRQ8-DS Pin Edge Select	0: A request for release is generated at a falling edge 1: A request for release is generated at a rising edge	R/W
b1	DIRQ9EG	IRQ9-DS Pin Edge Select	0: A request for release is generated at a falling edge 1: A request for release is generated at a rising edge	R/W
b2	DIRQ10EG	IRQ10-DS Pin Edge Select	0: A request for release is generated at a falling edge 1: A request for release is generated at a rising edge	R/W
b3	DIRQ11EG	IRQ11-DS Pin Edge Select	0: A request for release is generated at a falling edge 1: A request for release is generated at a rising edge	R/W
b4	DIRQ12EG	IRQ12-DS Pin Edge Select	0: A request for release is generated at a falling edge 1: A request for release is generated at a rising edge	R/W
b5	DIRQ13EG	IRQ13-DS Pin Edge Select	0: A request for release is generated at a falling edge 1: A request for release is generated at a rising edge	R/W
b6	DIRQ14EG	IRQ14-DS Pin Edge Select	0: A request for release is generated at a falling edge 1: A request for release is generated at a rising edge	R/W
b7	DIRQ15EG	IRQ15-DS Pin Edge Select	0: A request for release is generated at a falling edge 1: A request for release is generated at a rising edge	R/W

The DPSIEGR1 register is not initialized by the internal reset signal that is the source for release from deep software standby mode. For details, refer to Table 6.2, Targets to be Initialized by Each Reset Source.

11.2.19 Deep Standby Interrupt Edge Register 2 (DPSIEGR2)

Address(es): 0008 C28Ch

	b7	b6	b5	b4	b3	b2	b1	b0
	—	DRIICCEG	DRIICDEG	DNMIEG	—	—	DLVD2EG	DLVD1EG
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	DLVD1EG	LVD1 Edge Select	0: A request for release is generated when VCC < Vdet1 (fall) is detected 1: A request for release is generated when VCC ≥ Vdet1 (rise) is detected	R/W
b1	DLVD2EG	LVD2 Edge Select	0: A request for release is generated when VCC < Vdet2 (fall) is detected 1: A request for release is generated when VCC ≥ Vdet2 (rise) is detected	R/W
b3, b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b4	DNMIEG	NMI Edge Select	0: A request for release is generated at a falling edge 1: A request for release is generated at a rising edge	R/W
b5	DRIICDEG	SDA2-DS Edge Select	0: A request for release is generated at a falling edge 1: A request for release is generated at a rising edge	R/W
b6	DRIICCEG	SCL2-DS Edge Select	0: A request for release is generated at a falling edge 1: A request for release is generated at a rising edge	R/W
b7	—	Reserved	This bit is read as 0. The write value should be 0.	R/W

The DPSIEGR2 register is not initialized by the internal reset signal that is the source for release from deep software standby mode. For details, refer to Table 6.2, Targets to be Initialized by Each Reset Source.

11.2.20 Deep Standby Interrupt Edge Register 3 (DPSIEGR3)

Address(es): 0008 C28Dh

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	DCANIEG
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	DCANIEG	CRX1-DS Edge Select	0: A request for release is generated at a falling edge 1: A request for release is generated at a rising edge	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The DPSIEGR3 register is not initialized by the internal reset signal that is the source for release from deep software standby mode. For details, refer to Table 6.2, Targets to be Initialized by Each Reset Source.

11.2.21 Deep Standby Backup Register y (DPSBKRy) (y = 0 to 31)

Address(es): 0008 C2A0h to 0008 C2BFh

	b7	b6	b5	b4	b3	b2	b1	b0
Value after reset:	x	x	x	x	x	x	x	x

x: Undefined

The DPSBKRy register is a 32-byte readable/writable register to store data during deep software standby mode. The value of this register is retained even in deep software standby mode where RAM data is not retained. DPSBKRy is not initialized, and the register value is undefined immediately after power-on.

11.3 Reducing Power Consumption by Switching Clock Signals

When the SCKCR.FCK[3:0], ICK[3:0], BCK[3:0], PCKA[3:0], PCKB[3:0], PCKC[3:0], and PCKD[3:0] bits are set, the clock frequency changes. The CPU, DMAC, DTC, code flash memory, and RAM operate on the operating clock specified by the ICK[3:0] bits.

Peripheral modules operate on the operating clock specified by the PCKA[3:0], PCKB[3:0], PCKC[3:0], and PCKD[3:0] bits.

The external bus operates on the operating clock specified by the BCK[3:0] bits. For details, refer to section 9, Clock Generation Circuit.

11.4 Module-Stop Function

The module-stop function can be set for each on-chip peripheral module.

When the MSTPmi bit (m = A to D, i = 31 to 0) in registers MSTPCRA to MSTPCRD is set to 1, the specified module stops operating and enters the module-stop state, but the CPU continues to operate independently. When the corresponding MSTPmi bit is set to 0, the module is released from the module-stop state and restarts operating at the end of the bus cycle. The internal states of modules are retained in the module-stop state.

After release from a reset, all modules except for the DMAC, DTC, EXDMAC, RAM, ECCRAM, expansion RAM, and standby RAM are placed in the module-stop state. Though read/write access cannot be made to the registers of the module that are in the module-stop state, some registers may be written to directly after the setting to the module-stop state. Therefore, care should be paid.

11.5 Function for Lower Operating Power Consumption

By selecting an appropriate operating power consumption control mode according to the operating frequency and operating voltage, power consumption can be reduced in normal operation, sleep mode and all-module clock stop mode.

11.5.1 Setting Operating Power Consumption Control Mode

Examples of the procedures for switching operating power consumption control modes are shown below:

(1) Switching from Normal Power Consumption Mode to Low Power Consumption Mode

Example: From high-speed operating mode to low-speed operating mode 1

(High-speed operation in the operating power consumption control mode used before mode-switching)
↓
Set to switch from the HOCO clock to the LOCO clock (clock source and frequency division ratio)
↓
Write to register OPCCR (high-speed operating mode → low-speed operating mode 1)
↓
Confirm that the OPCCR.OPCMTSF flag is 0
↓
(Low-speed operation in the switched operating power consumption control mode)

(2) Switching from Low Power Consumption Mode to Normal Power Consumption Mode

Example: From low-speed operating mode 2 to high-speed operating mode

(Low-speed operation in the operating power consumption control mode used before mode-switching)
↓
Write to register OPCCR (low-speed operating mode 2 → high-speed operating mode)
↓
Confirm that the OPCCR.OPCMTSF flag is 0
↓
Set to switch from the LOCO clock to the HOCO clock (clock source and frequency division ratio)
↓
(High-speed operation in the switched operating power consumption control mode)

11.6 Low Power Consumption Modes

11.6.1 Sleep Mode

11.6.1.1 Transition to Sleep Mode

When the WAIT instruction is executed while the SBYCR.SSBY bit is 0, the CPU enters sleep mode. In sleep mode, the CPU stops operating but the contents of its internal registers are retained. Other peripheral functions do not stop.
When the WDT is used, the WDT stops counting when sleep mode is entered.
Counting by the IWDT stops if a transition to sleep mode is made while the IWDT is being used in auto-start mode and the OFS0.IWDTSLCSTP bit is 1 (stopping counting by the IWDT at transitions to low-power-consumption modes). In the same way, counting by the IWDT stops if a transition to sleep mode is made while the IWDT is being used in register start mode and the SLCSTP bit in IWDTCSTPR is 1.
Furthermore, counting by the IWDT continues if a transition to sleep mode is made while the IWDT is being used in auto-start mode and the OFS0.IWDTSLCSTP bit is 0 (counting by the IWDT continues through transitions to low-power-consumption modes). In the same way, counting by the IWDT continues if a transition to sleep mode is made while the IWDT is being used in register start mode and the SLCSTP bit in IWDTCSTPR is 0.
To use sleep mode, make the following settings and then execute a WAIT instruction.

(1) Set the PSW.I bit*1 of the CPU to 0.
(2) Set the interrupt request destination*2 to be used for recovery from sleep mode to the CPU.
(3) Set the priority*3 of the interrupt to be used for recovery from sleep mode to a level higher than the setting of the PSW.IPL[3:0] bits*1 of the CPU.
(4) Set the IERm.IENj bit*3 for that interrupt to 1.
(5) For the last I/O register to which writing proceeded, read the register to confirm that the value written has been reflected.
(6) Execute the WAIT instruction (this automatically sets the I bit*1 in the PSW of the CPU to 1).

Note 1. For details, refer to section 2, CPU.
Note 2. For details, refer to section 15.7.3, Selecting Interrupt Request Destination.
Note 3. For details, refer to section 15, Interrupt Controller (ICUD).

11.6.1.2 Release from Sleep Mode

Release from sleep mode is initiated by a non-maskable interrupt, an interrupt, the RES# pin reset, a power-on reset, a voltage monitoring reset, or a reset caused by an IWDT underflow.

- Release triggered by an interrupt signal
  Generation of an interrupt triggers release from sleep mode and the interrupt exception processing starts. If a maskable interrupt has been masked by the CPU (the priority level*1 of the interrupt has been set to a value lower than that of the PSW.IPL[3:0] bits*2 of the CPU), release from sleep mode does not proceed.
- Release due to a reset on the RES# pin
  When the RES# pin is driven low, the MCU enters the reset state. When the RES# pin is driven high after the reset signal is input for a predetermined time period, the CPU starts the reset exception processing.
- Release due to a power-on reset
  Release from sleep mode is initiated by a power-on reset.
- Release due to a voltage monitoring reset
  Release from sleep mode is initiated by a voltage monitoring reset from the voltage detection circuit.
- Release due to the independent watchdog timer reset
  Release from sleep mode is initiated by an internal reset generated by an IWDT underflow. However, when such conditions are set that stop IWDT counting in sleep mode (OFS0.IWDTSTRT = 0 and OFS0.IWDTSLCSTP = 1, or OFS0.IWDTSTRT = 1 and IWDTCSTPR.SLCSTP = 1), the IWDT is stopped and release from sleep mode is not initiated by the independent watchdog timer reset.

Note 1. For details, refer to section 15, Interrupt Controller (ICUD).
Note 2. For details, refer to section 2, CPU.

11.6.1.3 Sleep Mode Return Clock Source Switching Function

To switch the clock source used on return from sleep mode, the clock used after return needs to be set by the sleep mode return clock source switching register (RSTCKCR) and the wait control register needs to be set for each clock source. When the return interrupt is generated, after oscillation settling of the oscillator specified as the return clock, the clock source is automatically switched, and then operation returns from sleep mode. At this time, the registers related to clock source switching are automatically rewritten.
For details, refer to section 11.2.7, Sleep Mode Return Clock Source Switching Register (RSTCKCR). For setting a waiting time for oscillation stabilization, refer to section 9.2.17, Main Clock Oscillator Wait Control Register (MOSCWTCR).

11.6.2 All-Module Clock Stop Mode

11.6.2.1 Transition to All-Module Clock Stop Mode

After setting the MSTPCRA.ACSE bit to 1 and placing modules controlled by MSTPCRA, MSTPCRB, MSTPCRC, and MSTPCRD registers in the module-stop state (MSTPCRA = FFFF FF[C-F]Fh, MSTPCRB = FFFF FFFFh, MSTPCRC[31:16] = FFFFh, MSTPCRD = FFFF FFFFh), executing a WAIT instruction while the SBYCR.SSBY bit is 0 stops the bus controller, I/O ports, and all modules except for the 8-bit timers*1, POE*2, IWDT, RTC, power-on reset circuit, voltage detection circuit at the end of the current bus cycle, and the chip enters all-module clock stop mode*3.
When the WDT is used, the WDT stops counting when all-module clock stop mode is entered.
Counting by the IWDT stops if a transition to all-module clock-stop mode is made while the IWDT is being used in auto-start mode and the OFS0.IWDTSLCSTP bit is 1 (stopping counting by the IWDT at transitions to low-power-consumption modes). In the same way, counting by the IWDT stops if a transition to all-module clock-stop mode is made while the IWDT is being used in register start mode and the SLCSTP bit in IWDTCSTPR is 1.
Furthermore, counting by the IWDT continues if a transition to all-module clock-stop mode is made while the IWDT is being used in auto-start mode and the OFS0.IWDTSLCSTP bit is 0 (counting by the IWDT continues through transitions to low-power-consumption modes). In the same way, counting by the IWDT continues if a transition to all-module clock-stop mode is made while the IWDT is being used in register start mode and the SLCSTP bit in IWDTCSTPR is 0.
To use all-module clock-stop mode, make the following settings and then execute a WAIT instruction.

(1) Set the PSW.I bit*4 of the CPU to 0.
(2) Set the interrupt request destination*5 to be used for recovery from all-module clock stop mode to the CPU.
(3) Set the priority*6 of the interrupt to be used for recovery from all-module clock stop mode to a level higher than the setting of the PSW.IPL[3:0] bits*4 of the CPU.
(4) Set the IERm.IENj bit*6 for the interrupt to be used for recovery from all-module clock stop mode to 1.
(5) Read the last I/O register to have been written and confirm that its value reflects the value written.
(6) Execute a WAIT instruction (executing a WAIT instruction causes automatic setting of the PSW.I bit*4 of the CPU to 1).

Note 1. The MSTPCRA.MSTPA4 and MSTPA5 bits select operation or stop of these modules.
Note 2. When a POE interrupt source condition is satisfied while the setting to enable POE interrupts is in place, recovery from all-module clock stop mode does not proceed but the flag to indicate satisfaction of the source condition is retained. If a different source leads to recovery from all-module clock stop mode in this situation, a POE interrupt is generated after recovery.
Note 3. Transitions to all-module clock stop mode are not to be made in some states of DTC or DMAC operations. Before setting the MSTPCRA.MSTPA28 bit to 1, clear the DMAST.DMST bit of the DMAC and the DTCST.DTCST bit of the DTC so that the DTC and DMAC are not activated.
Note 4. For details, refer to section 2, CPU.
Note 5. For details, refer to section 15.7.3, Selecting Interrupt Request Destination.
Note 6. For details, refer to section 15, Interrupt Controller (ICUD).

11.6.2.2 Release from All-Module Clock Stop Mode

Release from all-module clock-stop mode is initiated by an external pin interrupt (the NMI or IRQ0 to IRQ15), a peripheral interrupt (8-bit timer*1, RTC alarm, RTC periodic, IWDT*2, USB suspend/resume, voltage monitoring 1, voltage monitoring 2, or oscillator-stopped detection interrupt), a RES# pin reset, a power-on reset, a voltage monitoring reset, or an independent watchdog timer reset, and the transition to the normal program execution state proceeds via exception processing for the given interrupt or reset.
However, note that in cases where a maskable interrupt has been masked by the CPU (priority level*3 of the interrupt has been set to a value lower than that of the PSW.IPL[3:0] bits*4 of the CPU) or a maskable interrupt has been set up as a trigger to start the DTC or DMA transfer, release from all-module clock stop mode will not proceed.

Note 1. The MSTPA4 and MSTPA5 bits of MSTPCRA select operation or stopping of these modules.
Note 2. If a condition for the independent watchdog timer to stop counting applied (OFS0.IWDTSTRT = 0 and OFS0.IWDTSLCSTP = 1, or OFS0.IWDTSTRT = 1 and IWDTCSTPR.SLCSTP = 1) at the time of a transition to all-module clock-stop mode, using a reset from the independent watchdog timer to release the chip from all-module clock-stop mode is impossible because the independent watchdog timer is stopped.
Note 3. For details, refer to section 15, Interrupt Controller (ICUD).
Note 4. For details, refer to section 2, CPU.

11.6.3 Software Standby Mode

11.6.3.1 Transition to Software Standby Mode

When a WAIT instruction is executed with the SBYCR.SSBY bit set to 1 and the DPSBYCR.DPSBY bit set to 0, a transition to software standby mode is made. In this mode, the CPU, on-chip peripheral functions, and the oscillator functions stop. However, the contents of the CPU internal registers, RAM data, on-chip peripheral functions, and the states of the I/O ports are retained. Whether the address bus and bus control signals are placed in the high-impedance or the output state is retained can be specified by the SBYCR.OPE bit. Software standby mode allows significant reduction in power consumption because the oscillator stops in this mode. Yet, the main and sub clock oscillators can be operated or stopped. For details, refer to Table 11.2, Entering and Exiting Low Power Consumption Modes and Operating States in Each Mode.

Set the DMAST.DMST bit of the DMAC and the DTCST.DTCST bit of the DTC to 0 before executing the WAIT instruction.

When the WDT is used, the WDT stops counting when software standby mode is entered because the oscillator stops. Counting by the IWDT stops if a transition to software standby mode is made while the IWDT is being used in auto-start mode and the OFS0.IWDTSLCSTP bit is 1 (stopping counting by the IWDT at transitions to low-power-consumption modes). In the same way, counting by the IWDT stops if a transition to software standby mode is made while the IWDT is being used in register start mode and the SLCSTP bit in IWDTCSTPR is 1.

Furthermore, counting by the IWDT continues if a transition to software standby mode is made while the IWDT is being used in auto-start mode and the OFS0.IWDTSLCSTP bit is 0 (counting by the IWDT continues through transitions to low-power-consumption modes). In the same way, counting by the IWDT continues if a transition to software standby mode is made while the IWDT is being used in register start mode and the SLCSTP bit in IWDTCSTPR is 0.

When the oscillation stop detection function is enabled (OSTDCR.OSTDE = 1), software standby mode cannot be entered. To make a transition to software standby mode, execute a WAIT instruction after disabling the oscillation stop detection function (OSTDCR.OSTDE = 0).

To use software standby mode, make the following settings and then execute a WAIT instruction.

(1) Set the PSW.I bit*1 of the CPU to 0.

(2) Set the interrupt request destination*2 to be used for recovery from software standby mode to the CPU.

(3) Set the priority*3 of the interrupt to be used for recovery from software standby mode to a level higher than the setting of the PSW.IPL[3:0] bits*1 of the CPU.

(4) Set the IERm.IENj bit*3 for the interrupt to be used for recovery from software standby mode to 1.

(5) For the last I/O register to which writing proceeded, read the register to confirm that the value written has been reflected.

(6) Execute a WAIT instruction (executing a WAIT instruction causes automatic setting of the PSW.I bit*1 of the CPU to 1).

Note 1. For details, refer to section 2, CPU.

Note 2. For details, refer to section 15.7.3, Selecting Interrupt Request Destination.

Note 3. For details, refer to section 15, Interrupt Controller (ICUD).

11.6.3.2 Release from Software Standby Mode

Release from software standby mode is initiated by an external pin interrupt (the NMI or IRQ0 to IRQ15), peripheral interrupts (the RTC alarm, RTC periodic, IWDT, USB suspend/resume, voltage monitoring 1, and voltage monitoring 2 interrupts), an RES# pin reset, a power-on reset, a voltage monitoring reset, or an independent watchdog timer reset.
When an interrupt initiates release from software standby, the oscillators which were stopped by the transition to software standby are restarted. After the oscillation of all these oscillators has become stable, operation returns from software standby.
Note that the oscillators are not stopped by the transition to software standby under the following two conditions, but the return from software standby still follows the period for stabilization of oscillation by the oscillators if they had been stopped.

- MOFCR.MOFXIN = 1 and MOSCCR.MOSTP = 0
- RCR3.RTCEN = 1 and SOSCCR.SOSTP = 0

(1) Release due to an interrupt
When an interrupt request from the NMI, IRQ0 to IRQ15, RTC alarm, RTC periodic, IWDT, USB suspend/resume, voltage monitoring 1, or voltage monitoring 2 interrupt is generated, each oscillator which was operating before a transition to software standby mode resumes oscillation. After the time for recovery from software standby mode has elapsed, the chip is released from software standby and starts interrupt exception processing.
The time for recovery from software standby mode is the oscillation stabilization waiting time plus the time required for operations by the software standby release sequencer.

tSBYi = tSBYOSCWT + tSBYSEQ
tSBYi (i = MC, EX, PC, PE, PH, SC, HO, LO): Recovery time from software standby mode
tSBYOSCWT: Oscillation stabilization waiting time
tSBYSEQ: Time required for operations by the software standby release sequencer

For the oscillation stabilization waiting time to be used in calculating the time for recovery from software standby mode, use the greatest value of the oscillation stabilization waiting time of the oscillators which are to be started. For the oscillation stabilization waiting times of the oscillators, refer to section 63, Electrical Characteristics.

(2) Release due to a reset on the RES# pin
Clock oscillation starts when the low level is applied to the RES# pin. Clock supply for the MCU starts at the same time. Keep the level on the RES# pin low over the time required for oscillation of the clocks to become stable. Reset exception processing starts when the high level is applied to the RES# pin.

(3) Release due to a power-on reset
Release from software standby mode proceeds after a fall in the power-supply voltage leads to the generation of a power-on reset.

(4) Release due to a voltage monitoring reset
Release from software standby mode proceeds after a fall in the power-supply voltage leads to the generation of a voltage monitoring reset.

(5) Release due to an independent watchdog timer reset
An internal reset due to an underflow of the IWDT leads to release from software standby mode.
However, if a condition for the independent watchdog timer to stop counting applied (OFS0.IWDTSTRT = 0 and OFS0.IWDTSLCSTP = 1, or OFS0.IWDTSTRT = 1 and IWDTCSTPR.SLCSTP = 1) at the time of a transition to software standby, using a reset from the independent watchdog timer to release the chip from software standby is impossible because the independent watchdog timer is stopped.

11.6.3.3 Example of Software Standby Mode Application

Figure 11.3 shows an example where a transition to software standby mode is made at the falling edge of the IRQn pin, and release from software standby mode is initiated at the rising edge of the IRQn pin.
In this example, an IRQn interrupt is accepted with the IRQCRi.IRQMD[1:0] bits of the ICU set to 01b (falling edge), and then the IRQCRi.IRQMD[1:0] bits are set to 10b (rising edge). After that, the SBYCR.SSBY bit is set to 1 and the WAIT instruction is executed. Thus a transition to software standby mode is made. After that, release from software standby mode is initiated at the rising edge of the IRQn pin.
To return from software standby mode, settings of the interrupt controller (ICU) are also necessary. For details, refer to section 15, Interrupt Controller (ICUD).

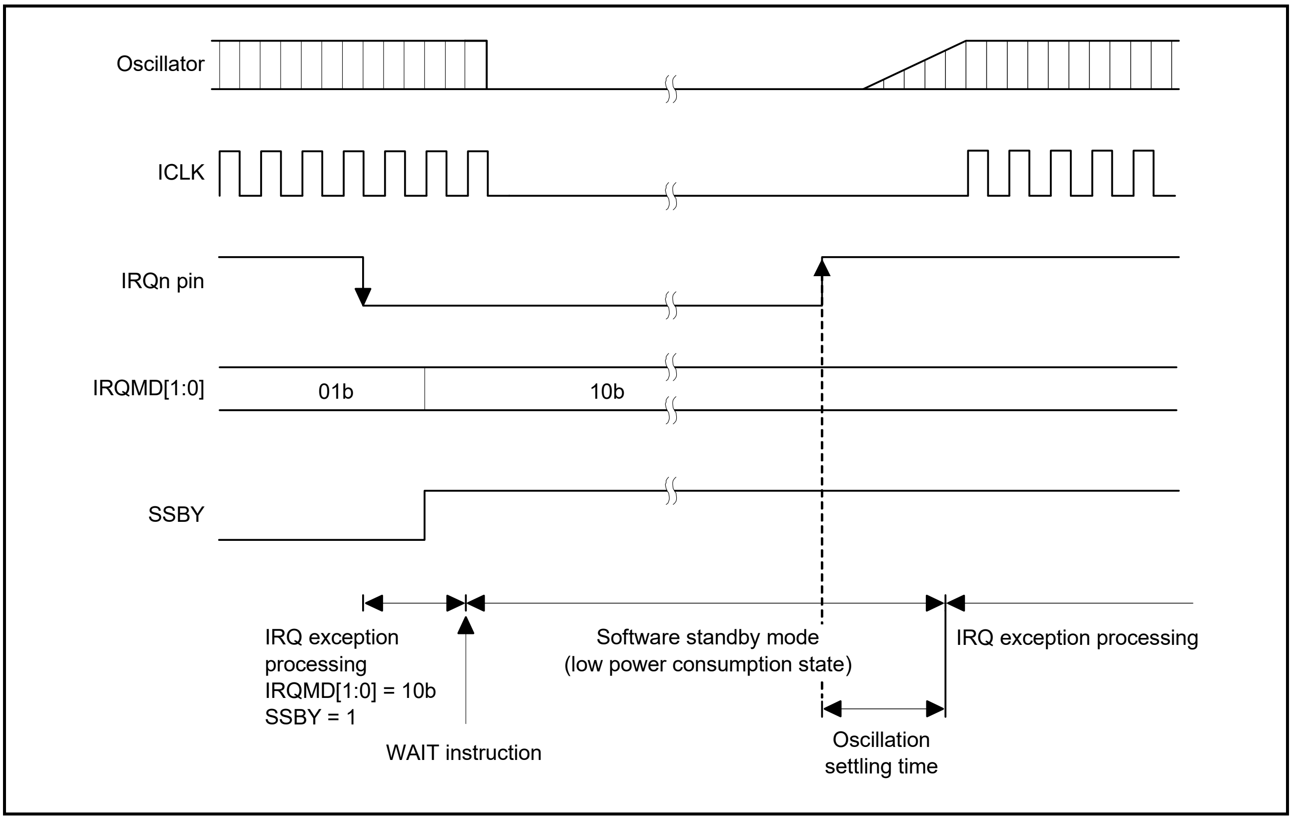


Figure 11.3 Example of Software Standby Mode Application

11.6.4 Deep Software Standby Mode

11.6.4.1 Transition to Deep Software Standby Mode

When the WAIT instruction is executed with the SBYCR.SSBY bit set to 1, a transition to software standby mode*1 is made. At this time, when the DPSBYCR.DPSBY bit is set to 1, a transition to deep software standby mode is made. On deep software standby mode, the CPU, internal peripheral modules (except for parts of the RTC alarm, RTC periodic, SCL2-DS, SDA2-DS, CRX1-DS, and USB suspend/resume detecting unit), RAM, expansion RAM, ECCRAM, and functions of the oscillators are stopped; furthermore, since the internal supply of power for these modules is stopped, power consumption is markedly reduced. Yet, the main and sub clock oscillators can be operated or stopped. For details, refer to Table 11.2, Entering and Exiting Low Power Consumption Modes and Operating States in Each Mode.
At this time, the contents of all the registers of the CPU and internal peripheral modules (except for parts of the RTC alarm, RTC periodic, SCL2-DS, SDA2-DS, CRX1-DS, and USB suspend/resume detecting unit) become undefined.
Data in the standby RAM are preserved if the setting of the DEEPCUT[1:0] bits is 00b. If the setting of the DEEPCUT[1:0] bits is 01b, the internal supply of power to the standby RAM and the USB resume detecting unit is cut off, reducing power consumption. Data in the standby RAM become undefined at this time. If the setting of the DEEPCUT[1:0] bits is 11b, the internal supply of power to the standby RAM, and the USB resume detecting unit is cut off, the LVD is stopped, and the low-power-consumption function of the power-on reset circuit is enabled, so power consumption is further reduced. For details, refer to section 63, Electrical Characteristics.
When the WDT is in use, since the oscillators and power supply to the WDT are stopped by the transition to deep software standby mode, counting also stops.
Power supply to the IWDT-dedicated clock and the IWDT is stopped and counting by the IWDT stops if a transition to deep software standby mode is made while the IWDT is being used in auto-start mode and the OFS0.IWDTSLCSTP bit is 1 (stopping counting by the IWDT at transitions to low-power-consumption modes). In the same way, power supply to the IWDT-dedicated clock and the IWDT is stopped and counting by the IWDT stops if a transition to deep software standby mode is made while the IWDT is being used in register start mode and the SLCSTP bit in IWDTCSTPR is 1.
Furthermore, counting by the IWDT continues if a transition to software standby mode is made but not to deep software standby mode while the IWDT is being used in auto-start mode and the OFS0.IWDTSLCSTP bit is 0 (counting by the IWDT continues through transitions to low-power-consumption modes). In the same way, counting by the IWDT continues if a transition to software standby mode is made but not to deep software standby mode while the IWDT is being used in register start mode and the SLCSTP bit in IWDTCSTPR is 0.
When the voltage monitoring 1 reset function (LVD1CR0.LVD1RI = 1) or voltage monitoring 2 reset function (LVD2CR0.LVD2RI = 1) is selected for the voltage detection circuit, a transition to deep software standby mode cannot be made, but to software standby.
The I/O port states remain unchanged from software standby mode.

Note 1. Conditions on the DTC, DMAC, and IWDT for transition to software standby mode should be met before the WAIT instruction is executed. For details, refer to section 11.6.3, Software Standby Mode.

11.6.4.2 Release from Deep Software Standby Mode

Release from deep software standby mode is initiated by any of the external pin interrupt source pins (the NMI, IRQ0-DS to IRQ15-DS, SCL2-DS, SDA2-DS, or CRX1-DS), peripheral interrupts (the RTC alarm, RTC periodic, USB suspend/resume*1, voltage monitoring 1, and voltage monitoring 2 interrupts), an RES# pin reset, a power-on reset, or a voltage monitoring 0 reset.

Note 1. USB is not released from deep software standby mode using USB0_OVRCURB multiplexed with pin P22.

(1) Release triggered by an external interrupt pin or internal interrupt signal
Release from deep software standby mode is controlled by registers DPSIERn (n = 0 to 3) and DPSIFRn (n = 0 to 3). When a deep software standby release interrupt is generated, the corresponding flag in DPSIFRn is set to 1. At this time, if the releasing source is enabled in DPSIERn, release from deep software standby mode proceeds. Rising edge or falling edge can be selected by DPSIEGRn register (n = 0 to 3). The interrupts for which an edge can be selected are the NMI, IRQ0-DS to IRQ15-DS, SCL2-DS, SDA2-DS, CRX1-DS, voltage monitoring 1, and voltage monitoring 2 interrupts.
When a deep software standby mode releasing source is generated, the internal power supply and LOCO clock oscillation begin, and then a deep software standby reset is generated for the entire MCU.
A stable LOCO clock is then supplied to the entire MCU, which is released from deep software standby reset. This is accompanied by release from deep software standby, and reset exception processing then starts.
When release from deep software standby is triggered by an external interrupt pin or internal interrupt signal, the RSTSR0.DPSRSTF flag is set to 1.

(2) Release due to a reset on the RES# pin
The low level being applied to the RES# pin triggers release from deep software standby.
At this time, the RES# pin should be held low according to the specifications described in section 63, Electrical Characteristics. Reset exception processing starts when the high level is applied to the RES# pin.

(3) Release due to a power-on reset
Release from deep software standby mode proceeds after a fall in the power-supply voltage leads to the generation of a power-on reset.

(4) Release due to a voltage monitoring 0 reset
Release from deep software standby mode proceeds after a fall in the power-supply voltage leads to the generation of a voltage monitoring 0 reset.

11.6.4.3 Pin States at the Time of Release from Deep Software Standby Mode

In deep software standby mode, the I/O ports retain the same states from software standby mode. The inside of the MCU is initialized by an internal reset generated on release from deep software standby mode. Upon release from deep software standby mode, the reset exception processing starts. The following shows the states of I/O ports at this time. Whether to initialize the I/O ports or to retain the I/O port states at the time of software standby mode can be selected by the DPSBYCR.IOKEEP bit.

- When the DPSBYCR.IOKEEP bit = 0
  I/O ports are initialized by an internal reset generated on release from deep software standby mode.
- When the DPSBYCR.IOKEEP bit = 1
  Although the inside of the MCU is initialized by an internal reset generated on release from deep software standby mode, I/O ports retain their states from software standby mode regardless of the MCU internal state. At this time, the I/O port states remain unchanged from software standby mode even if settings of I/O ports or peripheral modules are made. Then, the retained I/O port states are released by setting the DPSBYCR.IOKEEP bit to 0, and the MCU operates according to the internal state.

The DPSBYCR.IOKEEP bit is not initialized by an internal reset generated on release from deep software standby mode.

11.6.4.4 Example of Deep Software Standby Mode Application

Figure 11.4 shows an example where a transition to deep software standby mode is made at the falling edge of the IRQn-DS pin, and release from deep software standby mode is initiated at the rising edge of the IRQn-DS pin.
In this example, an IRQn interrupt is accepted with the IRQCRi.IRQMD[1:0] bits of the ICU set to 01b (falling edge). Then, after the DPSIEGRy.DIRQnEG (y = 0, 1; n = 0 to 15) bit is set to 1 (rising edge) and the SBYCR.SSBY bit and DPSBYCR.DPSBY bit are both set to 1, the WAIT instruction is executed. Thus a transition to deep software standby mode is made.
After that, release from deep software standby mode is initiated at the rising edge of the IRQ-DS pin.

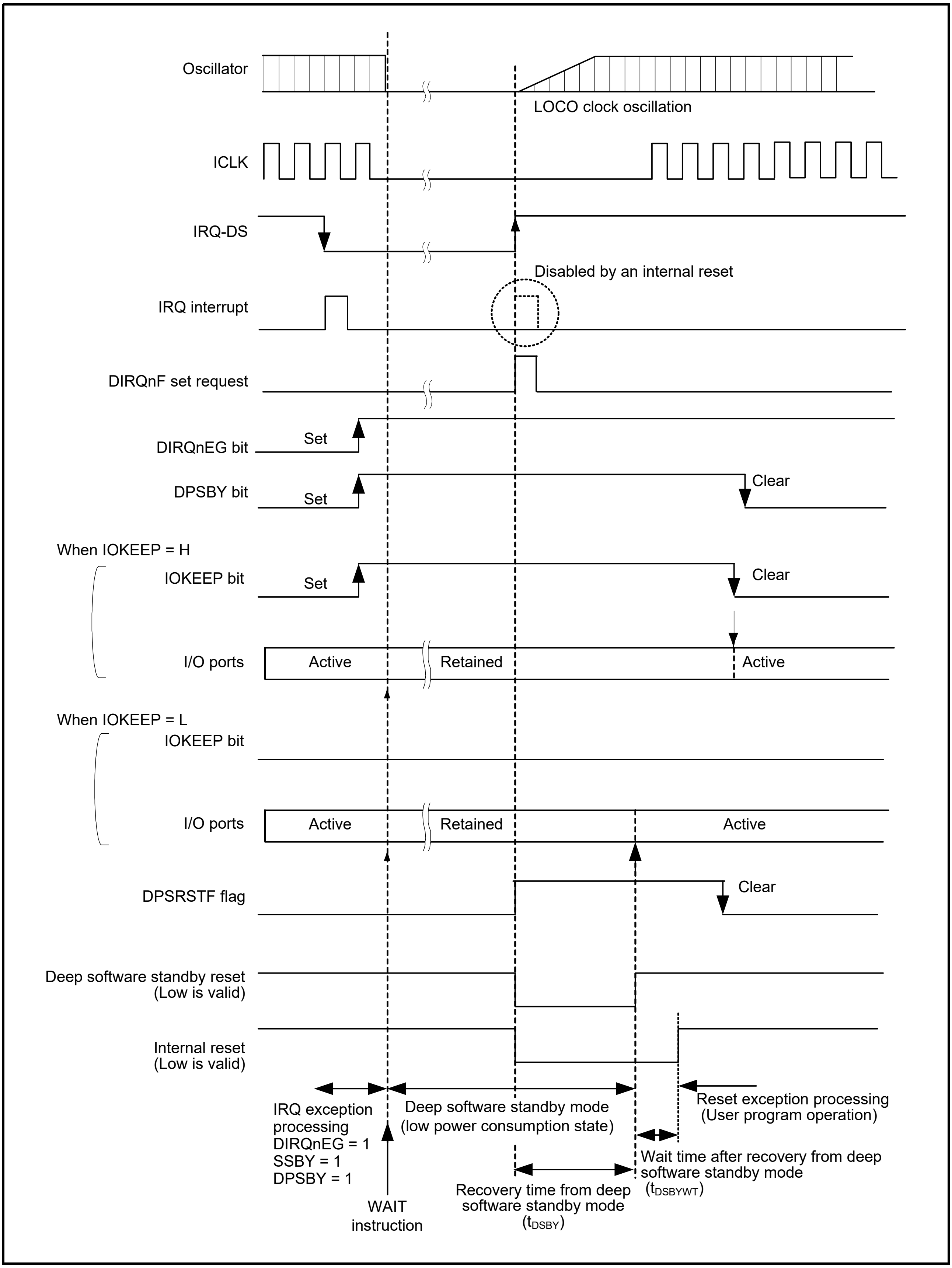


Figure 11.4 Example of Deep Software Standby Mode Application

11.6.4.5 Flowchart to Use Deep Software Standby Mode

Figure 11.5 shows an example of a flowchart to use deep software standby mode.
In this example, the RSTSR0.DPSRSTF flag of the reset function is read after the reset exception processing to determine whether a reset was generated by the RES# pin or by release from deep software standby mode.
In the case of a reset by the RES# pin, a transition to deep software standby mode is made after the required register settings have been made.
In the case of a reset by release from deep software standby mode, the DPSBYCR.IOKEEP bit is set to 0 after the I/O port settings have been made.

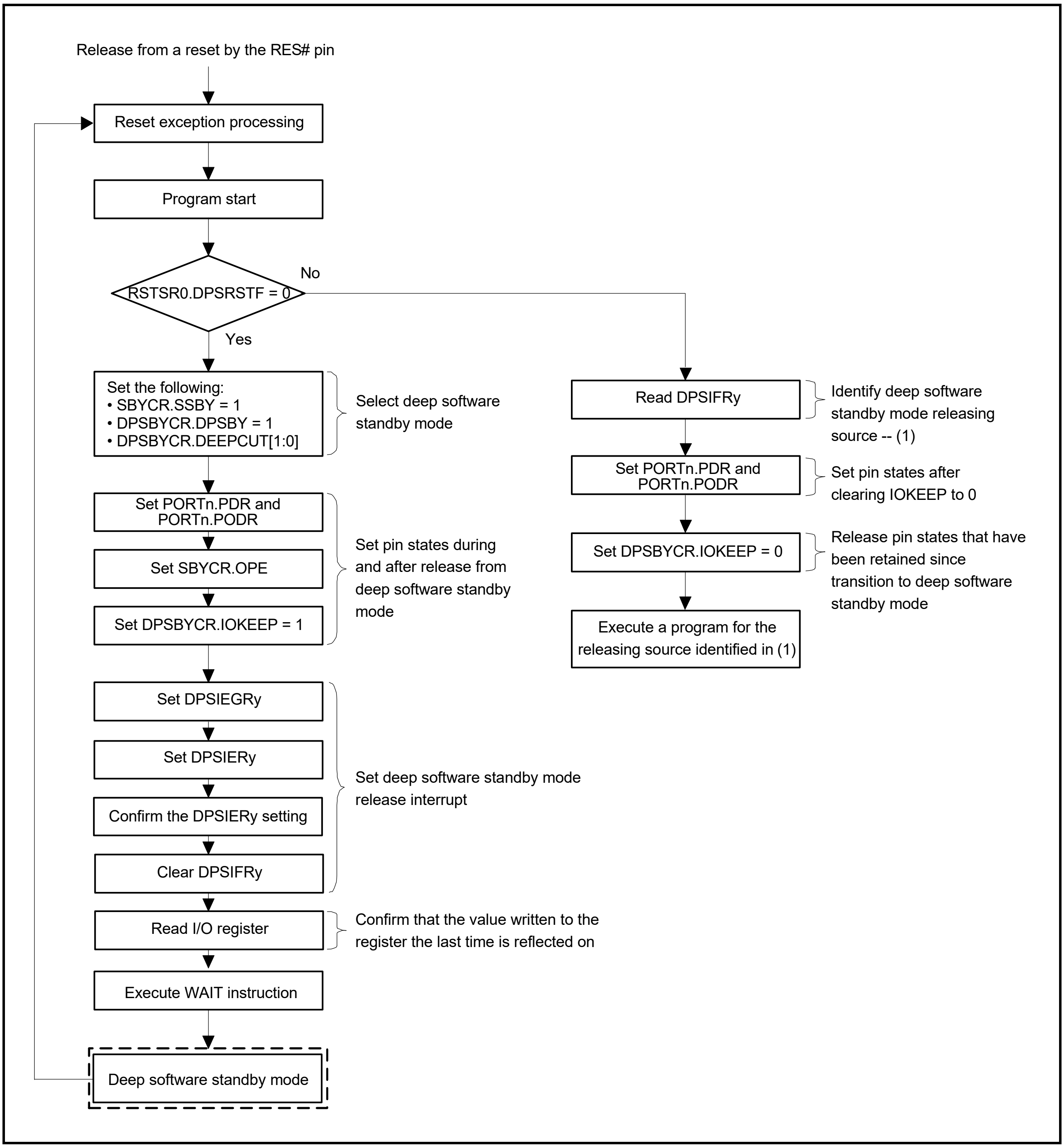


Figure 11.5 Example of Flowchart to Use Deep Software Standby Mode

11.7 Usage Notes

11.7.1 I/O Port States

I/O port states are retained in software standby mode and deep software standby mode. Therefore, the supply current is not reduced while output signals are held high.

11.7.2 Module-Stop State of DMAC and DTC

Before setting the MSTPCRA.MSTPA28 bit to 1, set the DMAST.DMST bit of the DMAC and the DTCST.DTCST bit of the DTC to 0 so that the DMAC and DTC are not activated.
For details, refer to section 18, DMA Controller (DMACAa) and section 20, Data Transfer Controller (DTCb).

11.7.3 On-Chip Peripheral Module Interrupts

These interrupts do not operate in the module-stop state. Therefore, if the module-stop state is entered after an interrupt request is generated, a CPU interrupt source or a DMAC or DTC startup source cannot be cleared. For this reason, disable interrupts before entering the module-stop state.

11.7.4 Write Access to MSTPCRA, MSTPCRB, MSTPCRC, and MSTPCRD

Write accesses to registers MSTPCRA, MSTPCRB, MSTPCRC and MSTPCRD should be made only by the CPU.

11.7.5 Input Buffer Control by DIRQnE Bit (n = 0 to 15)

Setting the DPSIERy.DIRQnE bit (y = 0, 1; n = 0 to 15) to 1 enables the input buffer of the IRQ0-DS to IRQ15-DS pins. Therefore, note that, although inputs to these pins are sent to the DPSIFRy.DIRQnF bit (y = 0, 1; n = 0 to 15), they are not sent to the interrupt controller, peripheral modules, and I/O ports.

11.7.6 Timing of WAIT Instructions

A WAIT instruction that follows register writing may be executed before the writing is completed. Accordingly, the WAIT instruction may be executed before the change to the setting of an I/O register is reflected, in which case operation may not be as intended. To avoid this, always execute the WAIT instruction after confirming that the last writing to the register has completed.

11.7.7 Rewriting the Register by DMAC and DTC in Sleep Mode

The WDT stops in sleep mode. Do not set up the DMACA and DTC to rewrite any registers related to the WDT while the chip is in sleep mode.
According to the settings of the OFS0.IWDTSLCSTP bit and IWDTCSTPR.SLCSTP bit, the IWDT may also stop in sleep mode. If that is the case, do not set up the DMAC and DTC to rewrite any registers related to the IWDT in sleep mode.
The RSTCKCR register can be set so that the clock source is switched on recovery from sleep mode. For this reason, rewriting the register while the chip is in sleep mode may lead to unintended operation, so do not allow rewriting of the RSTCKCR register in sleep mode.

11.7.8 Point for Caution when Shifting from Low-Speed Operating Mode to Software Standby Mode

On return from software standby, the chip enters high-speed operating mode. Even if a WAIT instruction is executed in low-speed operating mode, if generation of the return interrupt precedes completion of the transition to software standby and processing for the transition is canceled, the chip does not return to the mode before execution of the WAIT instruction. If this creates a problem, set the OPCCR.OPCM[2:0] bits to 000b during processing of the return interrupt.

11.7.9 Notes on Transitions from High-Speed Operating Mode to Low-Speed Operating Mode or to Low Power Consumption Mode

When the operating frequency of ICLK is 70 MHz or higher and the chip is to be shifted to low-speed operation, or is to be placed on software standby or deep software standby, change the frequency of ICLK to one quarter of its current frequency and then wait for 3 µs before initiating the desired transition.
Conversely, if the frequency of ICLK is to be changed to a frequency of 70 MHz or higher at the time of a transition from low-speed operation to high-speed operation, set the frequency to one quarter of the intended frequency and then wait for 3 µs before changing it to the intended frequency. For details, refer to section 9.10.8, Notes on Changing the ICLK Frequency.